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Quota baskets versus tax baskets in fisheries

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Abstract

Because managing stocks individually is costly, many fisheries bundle multiple stocks and manage them with a single instrument. I study when such bundling is desirable and whether it should take the form of a shared quota or a shared landing fee, extending the constant-escapement framework of Reed (1979) and Weitzman (2002) to a multispecies fishery under ecological uncertainty.

The fundamental condition for a well-designed basket is equalizing marginal profits across species, a condition that combines biological productivity, harvesting costs, and prices. Under this criterion, I find that the tax basket outperforms the quota basket. However, two qualifications refine this result: (i) both instruments coincide and are first-best when marginal profits are equalized at each species' optimal escapement in a deterministic setting; and (ii) as the number of species in the group increases, the quota's uncertainty loss is diversified away, reducing the average per-species welfare gap. Yet increasing the basket size has a second cost: the mismatch from pooling heterogeneous species grows with the number of species for both instruments. Larger baskets are not always better.

Bundling improves on species-specific management only when the monitoring costs it saves exceed this pooling loss.

JEL Classification: Q22, Q28, Q58, D81

1 Introduction

Allocating property rights through species-specific quotas has been shown to solve the common-pool resource problem in fisheries (Costello et al., 2016; FAO, 2025; Hilborn et al., 2020; Hoshino et al., 2020; Mangin et al., 2018). Without quotas, competing fishermen

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have the incentive to maximize their revenue by harvesting more today, depleting the available biomass tomorrow (Clark, 1990; Gordon, 1954). Despite the success of catch-shares, only rich countries can bear the costs of fully implementing such systems for many species (Mangin et al., 2018). In contrast, accepting that most commercial fisheries jointly harvest multiple species (Fulton et al., 2022; Spence et al., 2021), some countries pool many stocks and regulate them as one (Kulanujaree et al., 2020) or use an aggregated economic instrument for the whole group, a quota basket (Sanchirico et al., 2006). While quota baskets have historically been dismissed as a solution because they risk overfishing vulnerable species (Bonzon et al., 2013; Clark and Kirkwood, 1986), there is evidence that they are viable economically and ecologically (Collado et al., 2021; Fulton et al., 2022; Karr et al., 2021; Kritzer et al., 2023; Kulanujaree et al., 2020). There is a welfare trade-off between using a blunt instrument and reducing administrative costs. How do we reconcile the overfishing risk with the economic and ecological viability observed in practice?

Using species-specific regulation for multispecies fisheries poses challenges that stem from joint production, limited selectivity, imperfect information, and fishing gear choice (Branch, 2009; Christensen and Pauly, 2004; Copes, 1986; May and Clark, 1979; Nakamura, 2015; Pauly et al., 1998; Sanchirico et al., 2006). First, it ignores natural clusters (shoals) and species interactions through ecology, regulation, and non-selective fishing gear. A consequence is omitting the trade-off between sustainability and profit maximization caused by the choke species: the most restrictive species lead to premature season closure (Abbott and Wilen, 2009) and a shortage of quota that blocks the extraction of non-vulnerable species (Hatcher, 2022; Ulrich et al., 2017; Wo et al., 2022). This is the case of Canary and Yelloweye rockfish in California: the protection of the first forgoes revenue margins of the second, and a higher catch of the second endangers the rebuilding of the first. Hilborn et al. (2004) modeled managing these two species plus ten others in the West Coast fishery. They found that maximizing profits meant overfishing some species, but preventing overfishing in all species caused 90% yield loss. Mixed fisheries have repeatedly failed to achieve the double gains (stock health and intertemporal profit maximization) that characterize species-specific catch-share management.

Second, Tinbergen's (1952) rule would call for a corrective instrument for each failure, but managers are constrained by how many economic instruments they can credibly design, monitor, enforce, and govern. For fisheries, managers seek to correct common-pool resource failures. However, fully implementing regulation for even one species is data-intensive (Mangin et al., 2018; Su et al., 2020), costly to administer (Karr et al., 2021), and politically unstable (OECD, 2003). Even if the resources are available, species are hard to identify at port or during the fishing season (Pacific Fishery Management Council, 2024; Squires et al., 1998) or aggregated in the datasets (Su et al., 2020; Sun et al., 2023). Such problems are especially pronounced in Chinese fisheries, which often harvest a broad mix of species with minimal selectivity, leading to indiscriminate removal across trophic levels (X. Jin et al., 2020; Kritzer et al., 2023; Szuwalski et al., 2017). Low selectivity hampers control and monitoring efforts (Sun et al., 2023; Ye et al., 2013), while aggregation in official landings

data complicates identifying stock/harvest composition (Cao et al., 2017; Pauly and Froese, 2012). These issues limit the feasibility of traditional species-specific management that relies on precise stock assessments.

Third, using specific instruments for a group is not limited to task difficulty; it also extends to concrete harm resulting from the mismatch: it induces discarding of the low-value species[1] (Copes, 1986; Ulrich et al., 2017), effort reallocation toward unregulated stocks (Smith et al., 2009), and intertemporal arbitrage (Holzer and DePiper, 2019).

This aggregation problem belongs to a broader class of second-best regulation problems in which a single instrument must address multiple market failures. In environmental economics, this question has been studied for uniform versus differentiated taxes in a static setting across heterogeneous pollution sources (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986), and for the choice between prices, quantities, and mixed policies when multiple pollutants are jointly regulated (Ambec and Coria, 2013; Hoel and Karp, 2002). The fisheries literature also proposed alternatives for handling multiple stocks, such as a shared cap that is the percentage of the summation of individual caps (Sanchirico et al., 2006), a shared cap that targets a group mortality (Garcia et al., 2020), a shared cap with safeguards cap for vulnerable species (Sun et al., 2025), multispecies maximum sustainable yield (Fulton et al., 2022), aggregate production models (Fogarty et al., 2012), quota exchanges (Singh and Weninger, 2024), size-spectrum models (Andersen et al., 2016), among others. Most of these methods are helpful but prohibitive for many regulators in terms of scientific capability, information, enforceability, and budget. Moreover, many do not incorporate ecological uncertainty.

Unlike these solutions, quota baskets do not rely on a single-species framework or rich datasets (Collado et al., 2024). They consider full-bundling of the species under a single instrument, allowing flexibility in harvest allocation by trading stringency for convenience. However, the theoretical and empirical research on this subject is scarce. Clark's earliest theory (1990) showed that joint harvesting could not sustain both species in the short run, depleting the more vulnerable stock in a static open-access setting. Extending the analysis dynamically, he demonstrated that joint harvesting may be feasible but leads to an unstable equilibrium in an open-access fishery. In a non-fishery setting, Weitzman (1998) examined biodiversity preservation under budget constraints and showed that optimal bundling strategies may imply sacrificing some species to preserve others. Recently, Collado et al. (2021) analyzed the fisherman's choice of fishing gear under multiple basket allocations in a modified Gordon–Schaefer model. These results suggest that well-designed baskets can maintain moderate overall stock survival, whereas poorly designed configurations risk population collapse of the vulnerable species or all species in the group.

Finally, consensus remains lacking on how to group species or the conditions needed to achieve a welfare-maximizing quota basket, except in some empirical studies. For the first case, previous research recommends grouping species by similarity in ecological and

[1] From now on, high-grading

economic traits (Collado et al., 2021; Costello and Szuwalski, 2018), similar vulnerability indicators (catchability over intrinsic growth rate)(Collado et al., 2024), or relying on the expertise of local fishermen (EDF, 2020). Using empirical data from Belizean fisheries (United Nations Conference on Trade and Development, 2022), Collado et al. (2024) determined that the optimal quota basket was set at approximately 30% of the sum of individual maximum sustainable yields for selected Belizean baskets. These efforts build intuition but do not provide a definitive answer, especially in more complex settings such as managing ecological uncertainty.

In this paper, I define a tax basket that applies a single landing fee to a group of species. To my knowledge, no fishery in the world relies exclusively on landing fee regulations. Introducing the concept of a tax basket is useful because it lets me compare each instrument's capability to deal with ecological uncertainty. In this setting, I ask how much welfare the social planner gives up by using a basket instrument. Under what conditions do we need to establish a welfare-maximizing bundle instrument? If pooling species works, how does the choice between price and quantity matter under ecological uncertainty?

I use Reed's (1979) optimal remaining stock (escapement) as a reference for the species-specific double gains. This is the first-best outcome under a typical catch-share scheme. I also leverage Weitzman's (2002) information-flow framework to differentiate the information asymmetries on ecological uncertainty between fishermen and managers. With these pillars, I extend their analysis to a multispecies fishery for one regulator, one fisherman, and one basket. The model does not incorporate management costs, fishing gear, ecological dependence across species, or strategic interactions among fishermen to isolate the instrument's role under ecological uncertainty. For simplicity, I also assume that all species in the basket have commercial value for the fisherman. As a result, I do not cover bycatch or discarding.

I validate Weitzman's (2002) conclusion that tax baskets outperform quota baskets from an intertemporal welfare maximization goal. However, I find two qualifications that refine this result. First, if marginal profits under Reed's escapement are identical in a deterministic setting, quota and tax baskets have the same outcome and are first-best policies. Second, the per-species welfare gap between instruments narrows as more species are added to basket management. This diversification result links the basket design to portfolio theory (Edwards et al., 2004; D. Jin et al., 2016; Markowitz, 1952; Sanchirico et al., 2008), with the novelty that variance falls by adding species rather than optimizing portfolio weights. However, pooling loss diminishes diversification gains (the mismatch from forcing one instrument on heterogeneous species), and it grows with basket size. Under these circumstances, the policymaker must decide how to balance this trade-off with the savings of managing fewer administrative units.

To my knowledge, this is the first paper to study the incentives and trade-offs between a quota basket and a tax basket, which I introduce. I expand the seminal work of Reed (1979) and Weitzman (2002) to a multispecies setting.

This paper also speaks to multipollutant regulation literature (Ambec and Coria, 2013;

Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). I find that whether the manager uses a tax basket or a quota basket, fishermen will keep fishing each period until marginal profits per species are equalized. This result is the fisheries analog of the optimal uniform tax across heterogeneous pollutants, where the policy level is the weighted average of damages. While most multipollutant theory assumes a static, deterministic setting, this paper is set in a dynamic, stochastic setting.

The paper contributes to the literature on partial flexibility in multispecies quota systems. Where Singh and Weninger characterize the optimal partial flexibility parameter conditional on a species-specific fishing quota design, I derive the conditions for full bundling and compare the resulting quota basket with its tax counterpart. For how to bundle species, similarity across vulnerability indices is the criterion closest to the equimarginal principle presented in this paper. These indices are essentially a ratio of costs (catchability parameter) over revenue (intrinsic growth).

Finally, the paper contributes to the fisheries-under-uncertainty literature (Clark and Kirkwood, 1986; Hannesson and Kennedy, 2003; Reed, 1979; Sethi et al., 2005). The analysis shows that, even under uncertainty, basket instruments can be designed using a well-defined optimality criterion rather than ad hoc grouping rules.

The remainder of the paper is organized as follows. Section 2 presents the model and theoretical analysis. Section 3 reports the numerical results. Section 4 discusses policy implications and concludes.

2 Model

Throughout the paper, $t = 0, 1, \ldots$ denotes time, and $i = 1, 2, \ldots, N$ indexes the different species. The model assumptions rely on two cornerstones: (i) the intertemporal profits for a particular species are maximized with fixed escapement over time in deterministic and stochastic settings (Reed, 1979), and (ii) a fixed landing fee induces the fisherman to stop fishing at the optimal escapement [2](Weitzman, 2002). I model one basket, one fisherman, and one social planner or regulator.

For the ecology component, I take three key assumptions. First, I assume basket species' life cycles are independent, so the growth function depends only on its remaining stock y^i_{t-1} from the previous period. Then, the function follows the usual conventions: it is differentiable, concave $\frac{\partial^2 G(y^i_{t-1})}{\partial (y^i_{t-1})^2} \leq 0$, increasing $\frac{\partial G(y^i_{t-1})}{\partial y^i_{t-1}} > 0$ and bounded. Second, I assume the random ecological $\{Z^i_t\}$ are independent and identically distributed over time on some interval $[a, b]$ [3], with $\mathbb{E}[Z^i_t] = 1$ and $\mathbb{E}|\ln Z^i_t| < \infty$. Third, I assume weak exogeneity of stock on ecological shock and choice variables [4].

[2]An alternative proof can be found in Appendix A.

[3]The population is self-sustaining within the intervals $m = \max\{bG(s) : bG(s) \geq s, s \geq 0\}$ and $r = \max\{aG(s) : aG(s) \geq s, s \geq 0\}$

[4]The shock (Z^i_t) has zero conditional expectation given the present and past values of X^i_t and the choice

For fisherman's behavior, I make three key assumptions. First, the fisherman chooses the optimal fishing technology, allowing differentiation of the costs across species within a basket. Second, the prices are strictly positive for all species in the basket; so discards are not considered, as all harvest is valuable. Finally, all species can be fished in all periods (a fishable set $\mathcal{F}$ later defined in Lemma 2.1).

For the market component, I assume prices p^i, cost parameters θ^i, and the discount factor β are time-invariant. I also assume the marginal harvesting cost is decreasing in available stock s^i: $C^i(s^i) > 0$, $C'(s^i) < 0$. I work with the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, which denotes a constant catchability. I show that this constant-catchability form lets the tax deliver the manager's target escapements regardless of realized shocks.

I also adopt Weitzman's (2002) information framework, which combines a static optimization problem for a myopic fisherman with a forward-looking social planner:

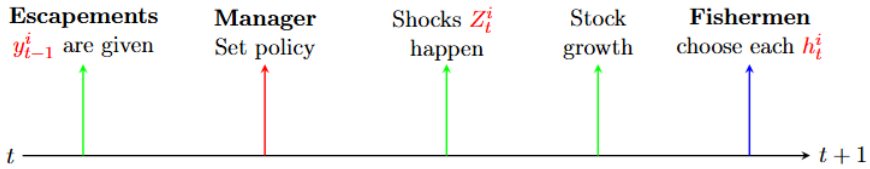


Figure 1: Information flow for each species i between periods t and $t+1$

At each period t, the following sequence of events occurs:

1. Escapements y^i_{t-1} for each species i are predetermined from the choices on the previous period.

2. The social planner sets a basket policy (a quota ω_t or a tax τ_t), unaware of how the stock composition will change with the ecological perturbations.

3. The ecological shocks Z^i_t are realized. Each species grows, and the stock shares change in the ocean.

4. The fisherman observes the realized stock and chooses the harvest h^i_t, given the basket policy set by the planner.

The key friction is that the planner chooses a basket policy to address multiple common-pool resource problems before shocks are realized. In contrast, the fisherman makes harvest decisions after shocks are realized. Given the economic instrument, a myopic fisherman observes the realized shocks and chooses harvest to maximize current profit:

1. Under a tax basket:

variables

$$\max_{\{h_t^i\}_{i=1}^N} \int_{X_t^i - h_t^i}^{X_t^i} (p^i - \tau_t - C^i(s^i))ds^i \tag{1}$$

2. Under a quota basket:

$$\begin{aligned} &\max_{\{h_t^i\}_{i=1}^N} \sum_{i=1}^N \int_{X_t^i - h_t^i}^{X_t^i} (p^i - C^i(s^i))ds^i \\ &\text{subject to} \quad \omega = \sum_{i=1}^N h_t^i \end{aligned} \tag{2}$$

where X_t^i is the stock at the start of period t, and h_t^i is the harvest. Due to the importance of the escapement y^i as a reference for species-specific profit maximization, I re-express the integral bounds using the escapement definition $y_t^i = X_t^i - h_t^i$:

$$\sum_{i=1}^N \int_{X_t^i - h_t^i}^{X_t^i} \pi^i(s^i)ds^i \to \sum_{i=1}^N \int_{y_t^i}^{X_t^i} \pi^i(s^i)ds^i$$

On the other hand, the forward-looking social planner sets the instrument (ω_t or τ_t) before shocks are realized to maximize discounted profits, taking the fisherman's decision rule for each species i as given:

1. Under a tax basket:

$$E\left[\sum_{t=0}^{\infty} \beta^t \sum_{i=0}^N \underbrace{\int_{X_t^i - h_t^i(\tau)}^{X_t^i} (p^i - \tau_t - C^i(s^i))ds^i}_{\text{Species i profit}} + \underbrace{\int_{X_t^i - h_t^i(\tau)}^{X_t^i} \tau_t ds^i(\tau)}_{\text{Species i tax revenue}}\right] \tag{3}$$

   Notice the tax terms in the profit and revenue components cancel.

2. Under a quota basket:

$$\begin{aligned} &E\left[\sum_{t=0}^{\infty} \beta^t \sum_{i=0}^N \int_{X_t^i - h_t^i(\omega)}^{X_t^i} (p^i - C^i(s^i))ds^i\right] \\ &\text{subject to} \quad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad \omega = \sum_{i=1}^N h_t^i \end{aligned} \tag{4}$$

where $\beta \in (0,1)$ is the discount factor. Assuming a risk-neutral government, the objective of this paper is the welfare for each instrument:

$$\boxed{\mathcal{W}^{\tau} - \mathcal{W}^{\omega}} \tag{5}$$

To calculate this difference, I obtain: (1). the fisherman's static decision rule for y_t^i, (2). the planner's dynamic optimized policy choice after incorporating (1), and (3). the fisherman's escapement response to the choice in (2). I use three benchmarks to describe the welfare results. The species-specific first-best, $\mathcal{W}^{\text{ind}}$, applies a separate optimal tax τ^{i*} to each stock. The tax basket, $\mathcal{W}^{\tau}$, applies a uniform landing fee τ^{+} to the group. The quota basket, $\mathcal{W}^{\omega}$, applies a single aggregate harvest limit ω_t. A policy is first-best if it attains $\mathcal{W}^{\text{ind}}$ and second-best if it maximizes intertemporal profits subject to the basket's constraints but yields strictly less: $\mathcal{W}^{\text{ind}} \geq \mathcal{W}^{\tau} \geq \mathcal{W}^{\omega}$.

Notice the model does not incorporate any assumption regarding the social planner's knowledge of the stock composition at the start of period t. This is deliberate because the Planner's problem section shows that the manager does not need to gather data on the initial biomass composition.

2.1 Fisherman's decision rule

From the FOC of Equation 3 concerning each harvest, I obtain the fisherman's optimality condition for the current period using the definition of escapement $y_t^i = X_t^i - h_t^i$:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = \tau_t \tag{6}$$

Notice that Equation 6 is the equimarginal principle: the total profits are maximized when the marginal profit of the last harvested unit for each species i is equal. The fisher sets the escapement decision:

$$\begin{aligned} p^i - \tau_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - \tau_t \\ y_t^i &= (C^i)^{-1}(p^i - \tau_t) \end{aligned} \tag{7}$$

For the static problem, notice that tax does not depend on the state variables (and I verify later in the planner's problem). In Equation 4, the fisherman faces the same maximization problem but for a given quota basket ω_t. The optimality condition in this case is:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = \lambda_t \tag{8}$$

Obtaining the same fisherman's escapement decision $y_t^i = (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t)$. Notice, however, that the shadow value is a function of the vector $\vec{X}$ and the quota basket ω_t because

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\ \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \lambda_t &= \lambda(\vec{X}_t, \omega_t) \end{aligned} \tag{9}$$

The fisherman decides their total harvest by equalizing the marginal profits $m\pi_t = \tau_t = \lambda_t(\vec{X}_t, \omega_t)$ across all species within the basket. Then, the fisherman's decision under a quota basket is:

$$y_t^i = (C^i)^{-1}(p^i - \lambda(\vec{X}_t, \omega_t)) \tag{10}$$

In this case, the shadow value depends on the state of all stocks, increasing its complexity relative to a tax basket. In any case, the key instrument to influence the optimality decision is the marginal profits, and tax and quota are vehicles to set them:

Lemma 2.1 (Fisherman's optimality condition)**.** *The optimality condition for harvesting N species in the fishable set $\mathcal{F} = \{i : p^i - m\pi > C^i(K^i)\}$ [5] is:*

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = m\pi_t$$

Through the stock-dependent cost function, the optimality condition incorporates all economic and biological parameters of all species.

The equimarginal principle holds: the total profits are maximized when the marginal profit $m\pi_t$ of the last harvested unit for each species i is equal.

2.2 Optimal marginal profits

The social planner incorporates the above fisherman's escapement decision into his dynamic optimization problem:

$$\begin{aligned} V_t(\vec{y_t}) = \max_{m\pi_t \geq 0} \ & \mathbb{E}\left\{ \sum_{i=1}^{N} \int_{y_t^i(m\pi_t)}^{Z_t^i G^i(y_{t-1}^i(m\pi_{t-1}))} \left(p^i - C^i(s^i)\right) ds^i + \beta V_{t+1}(\vec{X}_t) \right\} \\ \text{s.t. } & X_t^i = Z_t^i G^i(y_{t-1}^i) \end{aligned} \tag{11}$$

I take the FOC with respect to $m\pi_t$, and obtain the optimal marginal profit:

Proposition 1 (Welfare-maximizing $m\pi$)**.** *Consider N heterogeneous, independent stocks with cost function C^i, i.i.d. ecological shocks Z_t^i with $\mathbb{E}[Z_t^i] = 1$, and growth functions $G^i(y_{t-1}^i)$. Assume the interior regularity condition*

[5] Where K^i is the carrying capacity of species i

$$\max_i m\pi^{i*} \;<\; m\pi^F_{\max} \;:=\; \min_i\big(p^i - C^i(K^i)\big)$$

so that all species remain in the fishable set on $[\min_i m\pi^{i*}, \max_i m\pi^{i*}]$. *Then, the optimal* $m\pi_t^+$ *is determined by the following implicit function:*

$$m\pi_t^+ = \beta\frac{1}{\sum_{j=1}^N y^j_{m\pi_t^+}}\mathbb{E}\Big[\sum_{i=1}^N (p^i - C^i(Z_t^i G^i(y_t^i)))Z_t^i G^i_{y_t^i} y^i_{m\pi_t^+}\Big] \tag{12}$$

The implicit function above is the result of deriving the Bellman Equation and using the fisherman's condition $y_t^i = (C^i)^{-1}(p^i - m\pi_t)$ (see (38) in Appendix B.2). The Euler equation balances current marginal profit with the weighted average of next-period marginal profits.

Corollary 1.1 (The optimal $m\pi_t$ is constant)**.** *The optimal marginal profit is constant over time:* $m\pi_t = m\pi_{t+1} = m\pi$.

Proof. From expression 12, it is easy to observe that $m\pi \perp \vec{X}_t$. Because all species remain in the fishable set, the fisherman also equalizes marginal profits in period $t+1$:

$$\begin{aligned}
m\pi_t &= \sum_{i=1}^N \mathbb{E}\Big[m\pi_{t+1}\Big]\frac{y^i_{m\pi_t}}{\sum_{j=1}^N y^j_{m\pi_t}} \\
m\pi_t &= \sum_{i=1}^N \mathbb{E}\Big[m\pi_{t+1}\Big]\rho_t^i \\
m\pi_t &= \mathbb{E}\Big[m\pi_{t+1}\Big]\sum_{i=1}^N \rho_t^i \\
m\pi_t^+ &= \mathbb{E}\Big[m\pi_{t+1}\Big] \qquad [\text{because } \sum_i \rho_t^i = 1]
\end{aligned}$$

Because $X_t^i > y_t^i$ for $\forall i, t$, the summation of weights ρ_t^i is 1 because I assume all species remain in the fishable set. The Euler equation then equates current marginal profits with next-period expected marginal profits. On average, $m\pi_t$ is fixed over time. □

From this result, it is possible to obtain the following remarks:

Remark 1.1 (Deterministic case)**.** *If* $Z_t^i \equiv 1$, *then* $m\pi_t = m\pi_{t+1} = m\pi$.

Proof. With $Z_t^i \equiv 1$, the expectation in (12) drops, so $m\pi_t = m\pi_{t+1}$. □

Under a specific stock-dependent cost function, I can recover a marginal profit unaffected by the ecological perturbations.

Corollary 1.2 (Specific cost function recovers the deterministic case)**.** *If the cost function is* $C^i(s_t^i) = \frac{\theta^i}{s_t^i}$*, then* $m\pi_t = m\pi_{t+1} = m\pi$*.*

Proof. Let's replace $C^i(s_t^i) = \frac{\theta^i}{s_t^i}$ in Proposition 1:

$$\begin{aligned}
m\pi_t^+ &= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i - C^i(Z_t^i G^i(y_t^i))) Z_t^i G_{y_t^i}^i y_{m\pi_t}^i\Big] \\
&= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i - \frac{\theta^i}{Z_t^i G^i(y_t^i)}) Z_t^i G_{y_t^i}^i y_{m\pi_t}^i\Big] \\
&= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i Z_t^i - \frac{\theta^i Z_t^i}{Z_t^i G^i(y_t^i)}) G_{y_t^i}^i y_{m\pi_t}^i\Big] \\
&= \beta \frac{1}{\sum_{j=1}^N y_{m\pi_t}^j} \sum_{i=1}^N (p^i - \frac{\theta^i}{G^i(y_t^i)}) G_{y_t^i}^i y_{m\pi_t}^i \\
&= m\pi_{t+1} \sum_{i=1}^N \rho_t^i \\
&= m\pi_{t+1}
\end{aligned}$$

Under this cost function, the marginal profit behaves as in the deterministic case. If the cost function is $C^i(s_t^i) = \frac{\theta^i}{(s_t^i)^\alpha}$, then the marginal profit is a lower bound (see Appendix B). □

Remark 1.2 (Re-expression weights ρ_t^i)**.** *For a cost function* $C^i(s^i) = \frac{\theta^i}{s^i}$*, the weights* ρ_t^i *per species can be re-expressed as a function of its respective escapements and cost parameters:* $\rho^i = \frac{(y^i)^2/\theta^i}{\sum_j (y^j)^2/\theta^j}$

Proof. The re-expression of the weights is obtained by using the cost function $C^i(s^i)\frac{\theta^i}{s^i}$ and the fisherman's optimality condition $p^i - m\pi_t = C^i(y_t^i)$ in 12 (see expressions 39 and 40). □

By inspection, the tax basket equals $m\pi_t^+$, which does not depend on the state variables, while the quota includes the states through a stochastic shadow value. In consequence, the fisherman's escapement is deterministic under a tax ($\vec{\tau}$) and stochastic under a quota ($\vec{y}$). However, in a deterministic case, the shadow value is also deterministic because $\lambda(\vec{X}_t, \omega_t) = \lambda(Z_t^i \vec{G^i}(.), \omega_t) = \lambda(1\ \vec{G^i}(.), \omega_t)$. Then, the shadow value and tax basket behave the same way: $\tau_t = \lambda(\omega_t)$. In addition, if all species have the same marginal profit under their respective optimal escapements, then both instruments are first-best to target the optimal $m\pi^+$:

Corollary 1.3 (First-best basket instruments)**.** *Given the core assumptions, the tax basket or a quota basket implements a first-best $m\pi^+$ if and only if $p^i - C^i(y^{i*}) = p^j - C^j(y^{j*})$ for all $i \neq j$ in a deterministic setting.*

Proof. Note that if species have identical marginal profits at their respective optimal escapement levels, I can apply Reed's maximization $\frac{p^i - C^i(Z^i G^i(y^{i*}))}{p^i - C(y^{i*})} G'(y^{i*}) = \frac{1}{\beta}$ condition to each species to maximize profits over time:

$$\begin{aligned} p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = \tau_t = \lambda(\omega_t) \\ p^1 - C^1\left(y_t^{1*}\right) = p^2 - C^2\left(y_t^{2*}\right) = \cdots = p^n - C^n\left(y_t^{n*}\right) = \tau^* = \lambda(\omega_t) \\ \beta(p^1 - C^1(G^1(y^{1*}))G'(y^{1*}) = \cdots = \beta(p^n - C^n(G^n(y^{n*}))G'(y^{n*}) = \tau^* = \lambda(\omega_t) \end{aligned}$$

Even if species' profits are heterogeneous, a fixed tax or shadow value achieves Reed's constant escapement. □

This analysis is incomplete because I need to formally prove that $\tau^+ = m\pi_t^+$, and whether choosing the optimal shadow value is equivalent to choosing the optimal ω_t^+ to achieve the dynamically optimized $m\pi^+$.

2.3 Planner's problem

After studying the fisherman's decision and recognizing how the dynamic selection $m\pi^+$ can influence the harvest decision, the planner chooses the magnitude of the tax or quota basket to maximize intertemporal profits. For the tax basket, the solution of the dynamic problem is straightforward because it follows the same procedure as in Appendix B, after replacing $m\pi_t = \tau_t$. Because the tax is independent of the state variable in both the fisherman's and planner's problems, it holds the same optimality condition for the welfare-maximizing marginal profit of equation 38 in Proposition 1

Proposition 2 (Welfare-maximizing tax basket)**.** *Given the assumptions for N heterogeneous and independent stocks, species-specific optimal taxes τ^{i*} and interior regularity condition*

$$\max_i \tau^{i*} \; < \; \tau^F_{\max} \; := \; \min_i \big(p^i - C^i(K^i)\big),$$

[6]*, the welfare-maximizing tax basket τ^+ is characterized by the same implicit equation as in Proposition 1:*

$$\tau_t^+ = \beta \frac{1}{\sum_{j=1}^N y^j_{\tau_t^+}} \mathbb{E}\Big[\sum_{i=1}^N (p^i - C^i(Z_t^i G^i(y_t^i))) Z_t^i G^i_{y_t^i} y^i_{\tau_t^+}\Big] \tag{13}$$

[6]So that all species remain in the fishable set on $[\min_i \tau^{i*}, \max_i \tau^{i*}]$

Welfare is strictly lower than the summation of species-specific first-best whenever the τ^{i*} are not all equal. Then, τ^+ is the second-best. The tax basket τ^+ also satisfies corollaries 1.1 to 1.3.

Proof. By replacing $m\pi_t^+ = m\tau_t^+$ in Proposition 1 and all its related corollaries, it is easy to observe that the optimal tax basket mirrors the optimal marginal profit Euler equation and is fixed over time (for a complete derivation, see Appendix B.3). □

Notice that if I decompose the per-period profits:

$$\int_{y_t^i}^{X_t^i} \left(p^i - \frac{\theta^i}{s} \right) ds \;=\; p^i X - \theta^i \ln X - p^i y + \theta^i \ln y \tag{14}$$

and substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, using $\mathbb{E}[Z_t^i] = 1$ and $\mathbb{E}|\ln Z_t^i| < \infty$, it is possible to recover Proposition 2, showing that the same problem repeats each period, and it has the same solution even in a stochastic setting (more details in Appendix C). If the condition of Corollary 1.3 holds, then the tax basket is a first-best because all individual intertemporal optimal taxes are equal. Otherwise, by inspection, the tax basket is a second-best policy because it implements a constant escapement for all species that differs from their individual Reed escapement. It maximizes profits using a single instrument, but it does not achieve the same result of species-specific management.

Corollary 2.1 (Existence, uniqueness and interiority of τ^+)**.** *By Proposition 2, regularity condition* $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$ *and cost function* $C^i = \theta^i / s^i$*, and second-order condition*

$$y\, \pi_{t+1}^{i\prime}(y) + 2\pi_{t+1}^i(y) \;\leq\; p^i + \tau \tag{15}$$

The optimal tax basket exists, is unique, and is interior.

The full proof can be found in Appendix D. Equation 15 is a second-order condition that guarantees the global concavity and complements the fishable-set assumption to get a unique optimal tax basket. The fishable-set condition keeps every species harvested as τ varies, and 15 keeps each species' contribution concave. Both guarantee that a basket is not too heterogeneous, validating the similarity condition across species within the bundle (Collado et al., 2021, 2024; Costello and Szuwalski, 2018; Kritzer et al., 2023).

Taken together, these results show that tax baskets preserve the constant-escapement structure under uncertainty but generally implement a second-best allocation when species differ in marginal profits. First-best outcomes arise only when marginal profits are identical across species at their respective optima. For the policymaker having the data on economic and ecological parameters is enough to determine the optimal tax basket. However, data collection is necessary if $X^i > y^i(\tau^+)$ is not profitable for each i at every t, meaning that a species i can enter and leave the basket each period. In that case, the interiority condition breaks down.

In contrast, the quota basket requires more information because it changes over time. The essential difference with the tax basket is that a single quota constrains only aggregate harvest, so the split across species is the fisherman's choice; the shadow value λ_t is not a policy instrument but an equilibrium object that depends on the current stocks and the quota, $\lambda_t = \lambda(\mathbf{X}_t, \omega_t)$.

Proposition 3 (Welfare-maximizing quota basket)**.** *Under the setup of Proposition 2, the welfare-maximizing quota ω^+ set before the shocks are realized satisfies*

$$\mathbb{E}\Big[\lambda_t - \beta \sum_i \rho_t^i \, V_{y^i}(\vec{y}_t)\Big] = 0, \qquad \rho_t^i := \frac{Y_\lambda^i(\lambda_t)}{\sum_j Y_\lambda^j(\lambda_t)} = \frac{(y_t^i)^2/\theta^i}{\sum_j (y_t^j)^2/\theta^j}, \tag{16}$$

where $\lambda_t = \lambda(\mathbf{X}_t, \omega_t)$ is the fisherman's shadow value, and V is the planner's next-period value function, and

$$V_{y_{t-1}^k}(\vec{y}_{t-1}) = \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \tag{17}$$

The derivation is available in Appendix E.3. Equations 16 and 17 are the quota-equivalent of the tax first-order condition 13. Expression 16 equates, in expectation, the current shadow value of an extra unit of aggregate harvest with the ρ-weighted continuation value of the escapement. Equation 17 is the shadow value of one extra fish in stock k minus the λ_t of the escapement it displaces, plus its continuation value. Separating the optimality condition into two parts is useful because 16 helps to visualize the role of the ecological shocks, and 17 reveals what happens when there is a deviation between the shadow value chosen by the manager, λ_t^+, and the effective shadow value the fisherman selects after the shocks, λ_t^F. Before discussing how the planner's capacity to know the ecological shocks causes a mismatch across shadow values, I need to establish that choosing λ^+ is equivalent to choosing τ^+:

Corollary 3.1 (Duality of λ^+ and τ^+)**.** *Choosing the optimal quota ω_t^+ is equivalent to choosing an optimal shadow value λ^+, in the sense that the planner's first-order condition reduces to*

$$\lambda^+ = \beta \frac{1}{\sum_{j=1}^N y_{\lambda^+}^j} \mathbb{E}\Big[\sum_{i=1}^N (p^i - C^i(Z_t^i G^i(y_t^i))) Z_t^i G_{y_t^i}^i y_{\lambda^+}^i\Big] \tag{18}$$

which is identical in form to the tax condition (13)*. Both instruments target the same escapement vector $\bar{y}^i = \theta^i/(p^i - \tau^+)$.*

The equivalence is demonstrated in Appendix E. Thanks to duality, it is straightforward to establish that (i). λ^+ is fixed over time, (ii). λ^+ exists, is unique, and is interior under the same conditions as τ^+, and (iii). The quota ω_t is not constant. In other words, the two

instruments coincide at the planner's optimum but differ in the delivery. A regulator can determine the quota basket as the difference between the aggregated stock and the sum of optimal escapements:

$$\omega_t^+ = \sum_i X_t^i - \sum_i Y^i(\lambda^+) \tag{19}$$

As with a tax basket, determining a quota basket level does not require knowing the stock composition across species, but it does require knowing total biomass. By the end of period t, comparing a complete-information and information-asymmetry quota basket shows the difference between the fisherman's λ_t^F and planner's λ^+ shadow value:

$$\sum_i (Z_t^i - 1)\, G^i(y_{t-1}^i) = \sum_i \Big[(C^i)^{-1}(p^i - \lambda_t^F) - (C^i)^{-1}(p^i - \lambda^+) \Big], \tag{20}$$

so λ_t^F inherits the randomness of the aggregate realized stock. The following Corollary displays the source of distortion:

Corollary 3.2 (The distortion between τ^+ and ω_t)**.** *In a stochastic setting, the decomposition of the quota basket optimality condition*

$$\beta V_{y^k}(\vec{y}_{t-1}) = \pi^k(y_{t-1}^k) - \beta \operatorname{Cov}\big(R_t, Z_t^k\big)\, G^{k\prime}(y_{t-1}^k), \text{ where: } R_t = \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t) \tag{21}$$

where the distortion is the covariance between the residual across shadow values and the ecological shocks for each i.

The full proof is available at Appendix E.5. The covariance only vanishes in the deterministic case.

2.4 Welfare comparison

To compare the welfare outcomes across instruments, I borrow equation 41 (see the details in Appendix C.1)

$$\mathcal{W} = \sum_i \underbrace{[A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}]}_{=:\Xi_0} + \sum_{t\geq 0} \beta^t \sum_i \underbrace{[\phi^i(y_t^i) + \beta a^i(y_t^i)]}_{=:w^i(y_t^i)}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$ and $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$. Because Ξ_0 and the shock path are common to both instruments, I can cancel them in the difference. Then, comparing the per-species payoffs w^i:

$$w^{i\prime}(y) = \pi_{t+1}^i(y^i) - \big(p^i - C^i(y^i)\big), \qquad w^{i\prime\prime}(y) = \phi^{i\prime\prime}(y) + \beta a^{i\prime\prime}(y) < 0 \tag{22}$$

So $w^{i\prime}(y^i) = 0$ when $\pi^i(\bar{y}^i) = p^i - C^i(\bar{y}^i) = \tau^+$, generating the vector of escapements $\bar{y}$ that maximizes the intertemporal welfare. Under the tax, $y_t^i = Y^i(\tau^+) = \bar{y}^i$ is a fixed number, and the objective separates into a deterministic sequence of static problems for each i:

$$E[\mathcal{W}^\tau] = \Xi_0 + \frac{1}{1-\beta}\sum_i w^i(\bar{y}^i)$$

The quota does not separate. The escapement is random $y_t^i = Y^i(\lambda_t^F)$, function of every species' stock, with $\lambda_t^F = \lambda(\mathbf{X}_t, \omega_t^M)$ solving the extraction after the realization of the shocks Z_t^i. The vector of random escapements y_t is located in a random hyperplane $\mathcal{H}$. In contrast, the deterministic hyperplane $\mathcal{H}^* := \mathcal{H}_t\big|_{Z_t^i=1}$ corresponds to the tax basket's $\bar{y}$. Through duality, the planner sets ω_t to target $\mathcal{H}^*$, but the goal is not always reached:

Proposition 4 (The welfare is higher under a tax basket)**.** *Let both instruments implement the same optimal shadow value $\lambda^+ = \tau^+$ (Corollary 3.1). Under the interior regularity condition and i.i.d. shocks with $E[Z_t^i] = 1$,*

$$E[\mathcal{W}^\tau] \;\geq\; E[\mathcal{W}^\omega]$$

Proof. The tax delivers $\sum_i w^i(\bar{y}^i)$ while the quota delivers $E\big[\sum_i w^i(Y^i(\lambda_t^F))\big]$. Since Ξ_0 cancels, summing the discounted per-period gaps,

$$E[\mathcal{W}^\tau] - E[\mathcal{W}^\omega] = \sum_{t\geq 0}\beta^t\, E\Big[\sum_i \big(w^i(\bar{y}^i) - w^i(y_t^i)\big)\Big]$$

Because w^i is strictly concave with global maximizer $\bar{y}^i$ (22), Jensen's inequality gives

$$E\Big[\sum_i w^i\big(Y^i(\lambda_t^F)\big)\Big] \;\leq\; \sum_i w^i\big(E[Y^i(\lambda_t^F)]\big) \;\leq\; \sum_i w^i(\bar{y}^i),$$

Equivalently, by Appendix E.5, the tax attains $\beta V^\tau_{y^k} = \pi^k(y^k_{t-1})$ exactly while the quota attains $\beta V^\omega_{y^k} = \pi^k(y^k_{t-1}) - \beta\,\mathrm{Cov}(R_t, Z_t^k)G^{k\prime}$. Because λ_t^F is a continuous random variable, $\Pr(\lambda_t^F = \lambda^+) = 0$, so $\Pr(y_t^i = \bar{y}^i) = 0$. Hence $\mathcal{H}_t \neq \mathcal{H}^*$ almost surely. □

Under the assumptions of this paper, Weitzman's (2002) result holds: a tax outperforms a quota even under bundled management. This outcome rests on the cost function because the tax is stock-independent. Appendix B.3 shows that, under a cost function with decreasing catchability, the tax loses robustness and, in consequence, the ranking across instruments narrows. Also, the per-species welfare gap between instruments diminishes as more species are included:

Proposition 5 (Diversification of uncertainty loss as $N \to \infty$)**.** *The per-species welfare difference between instruments vanishes as $N \to \infty$.*

Proof. Under a quota basket, compare the escapement budget between the stochastic and deterministic settings: $\sum_i y_t^i = \sum_i Z_t^i G^i(y_{t-1}^i) - \omega_t$ and $\sum_i \bar{y}^i = \sum_i G^i(\bar{y}^i) - \omega_t$,

$$\sum_i (y_t^i - \bar{y}^i) = \sum_i [Z_t^i G^i(y_{t-1}^i) - G^i(\bar{y}^i)] = S_t, \qquad S_t \approx \sum_i (Z_t^i - 1) G^i(\bar{y}^i)$$

[7] if the scheme sits at the target. By variance additivity and independence,

$$\mathrm{Var}(S_t) = \sigma_Z^2 \sum_i [G^i(\bar{y}^i)]^2 = \Theta(N),$$

so the species-specific variance becomes $\mathrm{Var}(S_t/N) = \Theta(1/N) \to 0$ by the law of large numbers. Now, note that the escapements depend on the shadow values:

$$\sum_i \big[Y^i(\lambda_t^F) - Y^i(\lambda^+) \big] = S_t \tag{23}$$

Every Y^i is strictly increasing, so all N differences share the sign of $\lambda_t^F - \lambda^+$ and reinforce rather than cancel. Suppose $\lambda_t^F > \lambda^+$ (the reverse is symmetric). Then, by the mean value theorem, each difference is $Y^{i\prime}(\zeta^i)(\lambda_t^F - \lambda^+) \geq \underline{m}(\lambda_t^F - \lambda^+) > 0$ with the slope bound floor $\underline{m}$ of Lemma 4.2 (see Appendix F). After summing N deviations, I obtain:

$$S_t \geq \underline{m}\, N\, (\lambda_t^F - \lambda^+) \quad \Longrightarrow \quad |\lambda_t^F - \lambda^+| \leq \frac{|S_t|}{\underline{m}\, N}$$

Squaring and taking expectations,

$$E[(\lambda_t^F - \lambda^+)^2] \leq \frac{\mathrm{Var}(S_t)}{\underline{m}^2 N^2} = \frac{\Theta(N)}{\underline{m}^2 N^2} = \Theta(1/N) \to 0$$

Then, the shadow values converge, $E[(\lambda_t^F - \lambda^+)^2] = \Theta(1/N)$. Since $y_t^i = Y^i(\lambda_t^F)$ and $\bar{y}^i = Y^i(\lambda^+)$, the upper slope bound of Lemma 4.2 gives:

$$|y_t^i - \bar{y}^i| \leq \overline{m}^i\, |\lambda_t^F - \lambda^+| \quad \Longrightarrow \quad E[(y_t^i - \bar{y}^i)^2] \leq (\overline{m}^i)^2\, E[(\lambda_t^F - \lambda^+)^2] = O(1/N)$$

Because $\bar{y}^i$ is the basket maximum of w^i, Lemma 4.1 bounds the per-species loss by:

$$0 \leq w^i(\bar{y}^i) - w^i(y_t^i) \leq \tilde{B}^i\, (y_t^i - \bar{y}^i)^2, \qquad \tilde{B}^i := \tfrac{1}{2} \max_{[\underline{y}^i, \bar{y}^i]} \big(-w^{i\prime\prime} \big) < \infty$$

Taking expectations,

$$0 \leq w^i(\bar{y}^i) - E[w^i(y_t^i)] \leq \tilde{B}^i\, E[(y_t^i - \bar{y}^i)^2] = O(1/N)$$

[7] For simplicity, I evaluate at near target escapement

The per-species loss is $O(1/N)$. Summing the N per-species losses,

$$0 \leq E[\mathcal{W}^{\tau}] - E[\mathcal{W}^{\omega}] = \sum_i \big(w^i(\bar{y}^i) - E[w^i(y_t^i)]\big) \leq \sum_i \tilde{B}^i \, E[(y_t^i - \bar{y}^i)^2] = \sum_i O(1/N) = O(1)$$

Adding species diversifies each shock at rate $1/N$, but the total welfare loss persists.

□

Adding more species brings the quota basket's per-species outcomes closer to the tax basket. However, the gap does not disappear as $N \to \infty$.

3 Numerical results

To verify the welfare comparison, I simulate welfare and biomass under each instrument using the same randomly generated fisheries and Monte Carlo shock. I do not reproduce this analysis empirically to satisfy the regularity condition.

3.1 The dataset

Each replication draws a basket of N species satisfying the regularity condition under a cost function $C^i = \theta^i / s^i$:

$$\max_i \tau^{i*} \;<\; \tau^F_{\max} \;:=\; \min_i \Big(p^i - \frac{\theta^i}{K^i} \Big), \tag{24}$$

Drawing naively $(r^i, K^i, p^i, \theta^i)$ becomes challenging with increasing N because the rejection rate explodes. This corroborates that species cannot be too heterogeneous (Corollary 2.1). To avoid this, I generate the data by fixing escapement as a fraction of its carrying capacity $\iota^i := y^{i*}/K^i$ and inverting Reed's equation for the cost coefficient:

$$\theta^i \;=\; \frac{p^i \left(\frac{1}{\beta} - G^{i\prime}(y^i) \right)}{\dfrac{1}{\beta y^i} - \dfrac{G^{i\prime}(y^i)}{G^i(e)}}. \tag{25}$$

By holding p^i and ι^i fixed, the tax does not vary with K^i. With this, I draw the variables and then adjust θ^i. The ranges for my variables are modest enough that no single species dominates the aggregate. Consequently, I keep prices and escapement-fraction ranges deliberately narrow. Growth and carrying capacity capture the biological heterogeneity. Growth shocks Z^i_t are i.i.d. lognormal with unit mean, $Z^i_t = \exp(\mu + \sigma_Z \varepsilon^i_t)$, $\varepsilon^i_t \sim \mathcal{N}(0,1)$ and $\mu = -\sigma^2_Z/2$, so that $\mathbb{E}[Z^i_t] = 1$. Only the growth increment is perturbed: $x^i_{\text{pre}} = x^i + Z^i_t \, r^i x^i (1 - x^i/K^i)$.

3.2 Monte-Carlo simulation

For each N, I draw one basket and simulate both instruments over $T = 40$ periods on 6000 shock paths. Social welfare is the net present value of payoffs. For the tax, the social welfare also includes the net present tax revenue. I work with two variance controls: (i) common random numbers, where tax and quota face identical shocks; and (ii) antithetic variates such as ε and $-\varepsilon$. Together they reduce the standard error of the estimated welfare gap. My last check was verifying that the tax and quota yield identical welfare in a deterministic setting for heterogeneous baskets of size $N = 3, 17, 40$. I work with two basket scenarios: (1) a symmetric basket where every species is identical in parameters but different in perturbations, and (2) a fully heterogeneous basket.

For the decomposition of regulatory loss, I use two Monte Carlo layers: (1) the first layer averages over ecological shock paths holding a single basket of species fixed, and (2) a second layer that averages over random baskets. Hence, this simulation averages 25 baskets, each with 150 antithetic paths. I do this because I compare bundle- and species-specific instruments, and I need to smooth the latter's dispersion.

Table 1 summarizes the parameters and simulation considerations.

Parameter	Meaning	Range / value
r^i	intrinsic growth rate	$U[0.30, 0.40]$
K^i	carrying capacity	[1200, 2400]
p^i	price	$\mathcal{U}[95, 105]$
$\iota^i = y^{i*}/K^i$	escapement fraction	[0.48, 0.52]
θ^i	cost coefficient	from (25)
β	discount factor	$1/1.1 \approx 0.909$
σ_Z	shock volatility	0.20
T	horizon (periods)	40
N	basket size	$\{2, 10, 30, 50\}$
paths (Convergence)	Monte Carlo paths	6000
1st layer (Decomposition)	Baskets	25
2nd layer (Decomposition)	Monte Carlo paths	300

Table 1: Parameters of the simulated fisheries. Initial stocks are set to the target escapement, $X_0^i = \bar{y}^i$, so both instruments start from the same state and ensure all species are fishable.

3.3 Results

Panel (a) in Figure 2 shows that the expected net present welfare difference $\mathbb{E}[\mathcal{W}^\tau] - \mathbb{E}[\mathcal{W}^\omega]$ is positive at every N and converges to a constant difference: (i) around 209 dollars in the symmetric basket and (ii) 220–230 dollars in the heterogeneous scenario. In relative terms, the quota's loss is economically small (0.004% of $\approx$5.3 million) yet statistically significant (over 100 standard errors from zero). A quota does not outperform the tax basket, but increasing N drives the total gap to a positive and stable constant.

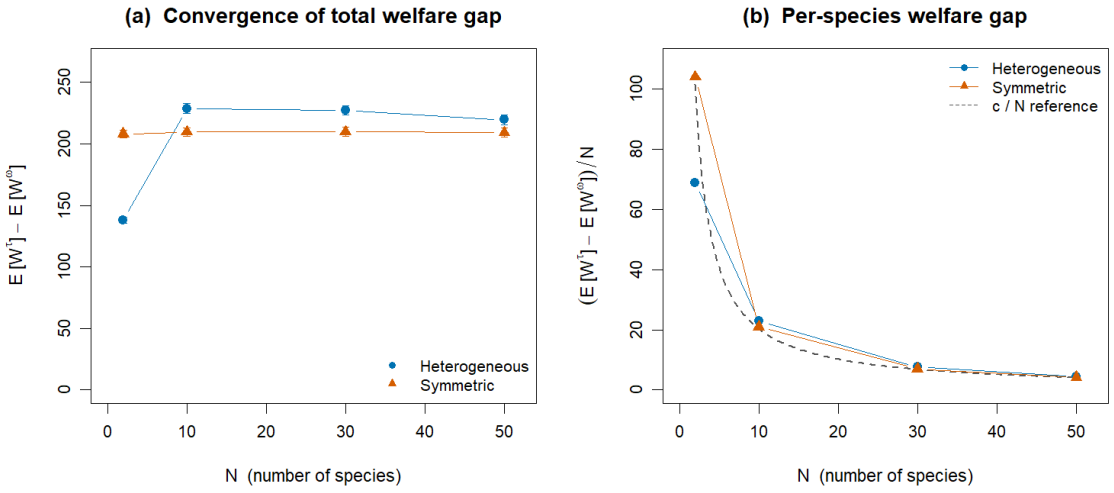


Figure 2: Left: expected total welfare difference $\mathbb{E}[\mathcal{W}^{\tau}] - \mathbb{E}[\mathcal{W}^{\omega}]$ per basket size N, the difference converges to a positive constant. Right: the per-species difference vanishes at rate $O(1/N)$.

Panel (b) divides the result of (a) by N: the per-species difference vanishes at rate $O(1/N)$, following the c/N reference curve. This simulation verifies that the quota's realized escapement vector converges in distribution to the tax's target $\bar{\bar{y}}$ as N grows. In addition, both the symmetric and heterogeneous scenarios benefit from increasing N, reducing the per-species gap. This helps reveal how uncertainty affects the welfare of a multispecies fishery.

3.3.1 Decomposing the regulatory loss

The difference between instruments shows the second-best cost of using shared quantity control instead of price control. To complete the comparison, I use the species-specific first-best taxes (τ^i*), which are the stringent instruments targeting each individual $m\pi^i*$. The sum of profits and tax revenues under this case generates the welfare W^{ind}. Because the first-best vector of taxes removes the losses associated with uncertainty and species pooling (forcing a shared instrument on heterogeneous individuals), I can order the welfare of each scheme as $W^{\text{ind}} \geq W^{\tau} \geq W^{\omega}$. With this consideration, I can decompose the regulatory loss:

$$\underbrace{\mathcal{W}^{\text{ind}} - \mathcal{W}^{\omega}}_{\text{1. total loss}} = \underbrace{\left(\mathcal{W}^{\text{ind}} - \mathcal{W}^{\tau}\right)}_{\text{2. pooling loss}} + \underbrace{\left(\mathcal{W}^{\tau} - \mathcal{W}^{\omega}\right)}_{\text{3. uncertainty loss}} \tag{26}$$

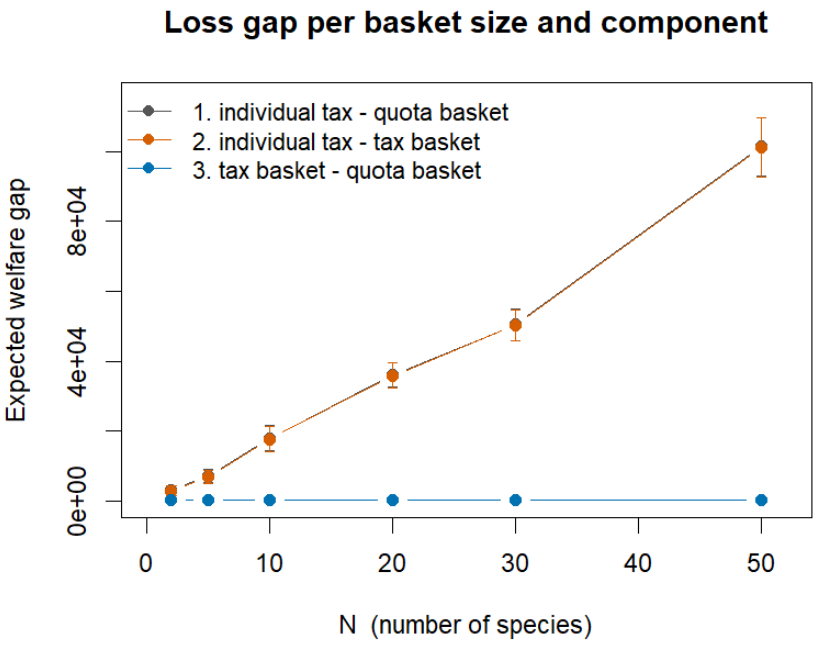


Figure 3: Decomposition of the total regulatory loss: 1. Expected total welfare loss difference between an individual tax and basket quota, 2. Expected total welfare loss difference between an individual tax and basket quota with N, while the 3. uncertainty loss is flat near zero.

Figure 3 shows components go in opposite directions in N. The uncertainty loss can be spread with additional species, reaching a stable loss of 217 dollars for all N. The per-species loss is 110 dollars at $N = 2$ to about 4 dollars at $N = 50$. The pooling loss goes in the opposite way. Its per-species welfare gap does not vanish: it stays bounded away from zero at roughly \$ 1,400–2,000 per species. Hence, the total pooling loss grows approximately linearly in N (from about \$ 2,900 at $N = 2$ to over \$ 10^5 at $N = 50$).

Using bundle instruments creates an additional tradeoff between instrument mismatch costs and uncertainty loss. For all N, the mismatch or pooling loss exceeds the uncertainty loss. This result extends previous research using groups of 2 (Clark, 1990), 5 (Collado et al., 2021), or 6 (Collado et al., 2024) species. This offers a lens on China's summer fishing moratorium, a seasonal closure applied to essentially all vessels and gear across the Bohai, Yellow, East, and South China Seas (Szuwalski et al., 2017). Rather than a tax or a species-specific quota, the moratorium is an aggregate effort control that manages the entire multispecies assemblage as a single basket: it reallocates no pressure across species and imposes no price or quantity controls. The decomposition suggests why such a coarse

instrument can still leave aggregate remaining biomass relatively stable, and why it must carry a large pooling loss in such a heterogeneous system.

3.3.2 Robustness

With the cost function $C^i(s^i) = \frac{\theta^i}{s^i}$, Remark 1.2 introduced per-species weights ρ_t^i as a function of their respective escapements and cost parameters. Under a tax basket, each species reaches a constant escapement, and its share of the next period's weighted-average profits becomes constant. In a quota basket, participation changes every period except when $N \to \infty$, because the quota approaches being fixed over time. In both cases, as more species are included in the basket, their weights shrink toward zero. I define a basket as balanced when the species whose share dominates the basket, $\max_i \rho^i$, also shrinks, $\max_i \rho^i \to 0$ as $N \to \infty$. What happens if a species' participation dominates because it is highly valuable and abundant? Because the share does not vanish, $\alpha^i \to \bar{\alpha} > 0$, its escapement variance [8] $\mathrm{Var}(y_t^i) = (\alpha^i)^2 \mathrm{Var}(S_t) = \bar{\alpha}^2 \Theta(N)$ grows with N. In other words, this "fixed" share makes it hard to distribute shocks across species.

To test this, I hold the share of one species in aggregate biomass (60%) fixed as N grows, so its idiosyncratic shock never diversifies away. Then I repeat the previous simulation (using the first layer for the Monte Carlo) with the dominant species. Because the shocks never diversify, the per-species difference no longer vanishes (it stays near 125–240), and the total welfare difference grows with N (from about 250 at $N = 2$ to over 12,000 at $N = 50$). Figure 4(a) shows that the loss is greater when diversification fails.

This robustness check shows that increasing the number of fishable stocks is not enough to reduce the per-species welfare gap. This case shows that bundle fisheries do not work for all species. A successful basket is a balanced one because it guarantees diversification. Relaxing the constant catchability in the selected cost functional form blurs the tax basket's advantage over the quota but leaves the pooling loss intact.

[8] I define the fraction of an aggregate-quota shock absorbed by species i as $y_t^i - \bar{y}^i \approx \alpha^i S_t$.

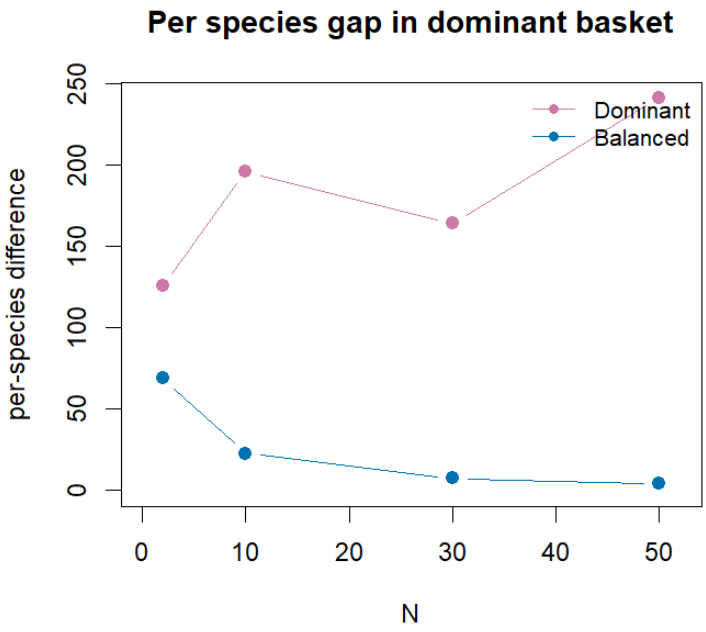


Figure 4: Robustness with one persistently dominant species. Per-species welfare does not vanish when one species' participation dominates, even as basket size increases, relative to a balanced basket.

4 Discussion and conclusion

This is the first paper to study the incentives and trade-offs for basket instruments for heterogeneous stocks under ecological uncertainty. By building on and extending the seminal works of Reed (1979) and Weitzman (2002), I find that tax baskets outperform quota baskets from a welfare standpoint. However, the per-species welfare gap narrows across instruments as more species are included in the bundle. As the basket size increases, a trade-off emerges between reducing ecological uncertainty and losing aggregate performance (relative to a first-best policy). Whether the policymaker uses a tax basket or a quota basket, fishermen will keep fishing each period until marginal profits are equalized across all fishable species. This margin integrates biological productivity, harvesting costs, and market prices. These findings help unify literature that proposes grouping species based on similar ecological (Costello and Szuwalski, 2018), economic (Collado et al., 2021), or vulnerability (Collado et al., 2024) parameters.

In general, both policies are second-best instruments, with tax baskets outperforming quota baskets in welfare. Both instruments are a weighted average of the marginal profits of

each species, analogous to how uniform taxes are calculated for multi-pollutants (Diamond, 1973; Fowlie and Muller, 2019; Mendelsohn, 1986). Because the problem is dynamic, the shares can change every period. In this context, a tax basket is particularly helpful because it fixes the remaining biomass (and shares) over time and is impervious to ecological shocks. While this result is attractive, the tax basket loses effectiveness relative to the species-specific instrument as N increases because a uniform tax on heterogeneous species reduces the instrument's stringency. Notwithstanding, a tax basket can achieve the outcomes of a first-best instrument under specific conditions. One condition is that all species are identical or have similar economic and biological parameters, as Kritzer et al. (2023) recommend. The most relevant condition is that each species' marginal profits under its respective Reed's escapement are identical (in other words, each individual optimal tax is the same). I show there is a duality between instruments, and this second recommendation also applies to quota baskets in a deterministic setting.

Quota baskets, by contrast, inherit an additional distortion under uncertainty because ecological shocks diminish their effectiveness. The timing asymmetry between manager and fisherman does not affect the tax basket because it automatically induces the fisherman to stop harvesting when the stock is scarce. However, when the quota basket includes many species and ecological shocks are independent, aggregation causes shocks to average out, reducing species-specific loss relative to the bundle tax. In this case, quota baskets can approximate the second-best tax, and the total welfare gap converges as N increases. Under the model's assumptions, prices still dominate quotas, but specific conditions can close the gap. Finally, the quota basket also performs worse relative to the stringent instruments as basket size increases.

The trade-off between uncertainty diversification and policy mismatch motivates the policymaker to choose the right N. In the simulation, I show that the tax outperforms the quota by only a fraction of a percent, while the pooling loss reaches six figures at $N = 50$. While this paper does not determine the optimal basket size, it shows that composition matters. First, the planner must decide whether the pooling loss justifies the system's administrative costs. If basket size is large and no monitoring capabilities exist, a bundle instrument would be recommendable as long as the total welfare loss does not exceed the regulatory expenses. A larger basket is not always better.

On the other hand, if institutional and scientific capabilities are well developed, a large basket may not be a priority because monitoring one additional species can be relatively cheap (if the number of monitored species is already high). This may explain the stark differences in "basket" experiences across fisheries (California, China, Alaska, Chile and Thailand). When datasets are hard to build, species are hard for managers to distinguish, or fishing gear is not selective enough, a basket instrument can be attractive. Basket instruments can complement existing super-species models (pooling all species as a single stock) used in Southeast Asia (Kulanujaree et al., 2020), especially for dealing with concerns about system-wide shifts such as climate change (Fulton et al., 2022).

These findings are particularly relevant for fisheries characterized by low selectivity, lim-

ited data, and high landing aggregation. For example, China harvests many species jointly, and regulators have limited ability to identify or regulate individual stocks (Szuwalski et al., 2017). In this setting, the moratorium functions as a whole-fishery effort control (the most aggregated basket possible), which these results predict will smooth ecological variability across stocks while incurring a pooling loss that rises with the number of species swept into it. This logic aligns with broader evidence that diversification across multiple stocks can reduce income volatility and targeting risk for fishers (Cline et al., 2017; Robinson et al., 2020). In contrast, aggregate production models have proven tractable in fisheries such as Thailand, but doubts remain about their resilience under ecological uncertainty (Fulton et al., 2022). Conceptually, the result parallels findings in environmental economics showing that uniform instruments can perform well when uncertainty is diversified across regulated sources (Fowlie and Muller, 2019).

Few countries employ landing fees, and where they exist, they typically complement rather than replace quota systems (Ben-Hasan et al., 2021), despite Weitzman's (2002) case for prices over quantities. This paper does not overturn that ranking: under ecological uncertainty the tax basket remains the welfare-preferred instrument. Yet if tax baskets are politically infeasible, this paper provides conditions under which a quota basket can narrow the welfare gap.

For the policymaker, basket instruments are most attractive in data-limited settings where species-specific regulation is infeasible. Tax baskets do not require real-time stock-level information at the start or during period t. While they reduce monitoring and species-identification costs, harvest-cost parameters are not readily available for all fisheries and may be difficult for authorities to obtain. Through instrument duality, quota baskets benefit from aggregation but require periodic updates and reliable aggregate biomass information. This weakens the presumption that basket regulation reduces monitoring costs. While indicator species may partially alleviate this burden, they introduce additional measurement error that compounds ecological uncertainty and further degrades quota performance [9]. These distinctions suggest that the choice between instruments should depend not only on ecological uncertainty but also on the regulator's institutional and budgetary capacity.

Several limitations point to directions for future research. First, the analysis abstracts from fishing technology choice and gear selectivity. Incorporating endogenous targeting decisions could clarify when basket instruments induce efficiency losses or gains through technological adjustment. Second, the framework focuses on a single basket; extending it to multiple, overlapping baskets raises questions about species mobility across regulatory groups and enforcement. Third, although the model does not explicitly represent species interactions, preliminary derivations suggest the result is ambiguous. Fourth, linking the marginal-profit condition to trait-based or size-spectrum ecosystem models (Andersen et al., 2016) could help translate theoretical criteria into empirically observable metrics, informing

[9] Sethi et al. (2005) demonstrated that measurement uncertainty has the greatest impact on fishery policy, profits, and fishery collapse under single-species management.

practical basket design that balances efficiency, biodiversity, and risk. Fifth, some results rest on the form of the cost function, $C^i(s^i) = \frac{\theta^i}{s^i}$, which makes the tax basket stock-independent. Relaxing to a non-constant catchability would narrow the tax's advantage.

Finally, this work does not consider bycatch of non-valuable or endangered species (e.g., Canary and Yelloweye rockfish in California). A bundle instrument could delay recovery of vulnerable species, but it would not heavily restrict extraction of healthier stocks. However, a non-healthy stock combined with our differentiated cost and fishable set assumption is problematic. Under differentiated costs, the fisherman decides what to catch, and the regularity condition is broken if a stock is scarce. If species enter and exit the fishable set, the solution adjusts the tax basket for the participating species to reduce welfare loss. In this case, the tax basket and species-specific escapements are not fixed. Moreover, in this scenario, the manager must invest in gathering information on participating species, and ecological shocks may change species participation after the policy is set. This creates an unplanned catch and a trade-off between conservation (designing the policy to include all species) and profit maximization (designing the policy to include only potentially fishable species). In short, bundle instruments cannot solve the sustainability-profitability trade-off they were created to address.

In summary, tax baskets are generally more robust under uncertainty, and both instruments can perform well in highly aggregated systems or under restrictive conditions. However, managing larger baskets implies a trade-off between diversifying uncertainty loss, increasing instrument-mismatch loss, and reducing administrative costs. Bundling is not a substitute for protecting vulnerable, non-target stocks. But, with careful attention to information requirements and ecological risk, basket regulation offers a pragmatic complement to species-specific management. Finally, implementing a basket instrument can be a stepping stone toward more stringent regulation schemes.

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Appendix A: Derivation of the optimal tax basket (1 species)

A.1: Fisherman's problem

The myopic fisherman maximizes profits through harvesting fish in period t:

$$\pi_t(x) = \max_{h_t \geq 0} \left\{ \int_{x_t - h_t}^{x_t} (p - \tau_t - C(s))\, ds \right\} \tag{27}$$

Where p is the unit price of fish, $C(s)$ represents the unit cost of harvesting (at fish population s), and τ is the landing fee per fish unit. After taking the first-order condition (FOC) concerning harvest, we obtain the optimality condition for the current period:

$$p - C\left(x_t - h_t\right) = \tau_t$$

Using the definition of escapement $y_t = x_t - h_t$, we can re-express this condition as:

$$p - C\left(y_t\right) = \tau_t \tag{28}$$

Notice that the fisherman takes an escapement decision:

$$\begin{aligned} p - \tau_t - C(y_t) &= 0 \\ C(y_t) &= p - \tau_t \\ y_t &= C^{-1}(p - \tau_t) \end{aligned} \tag{29}$$

A.2: Social planner's problem

The social planner tries to pick a landing fee to induce harvest levels that will maximize expected present-discounted profits, having the following form:

$$E[\sum_{t=0}^{\infty} \beta^t \int_{X_t - h_t}^{X_t} \pi(s) ds] \tag{30}$$

where $\beta \in (0,1) \equiv \frac{1}{1+r}$ and the recruitment of fish stock during period t is denoted by $X_t = Z_t G(y_{t-1})$

A.3: Landing fee

The manager sets τ_t considering the fisherman decision function and the previous end-of-period escapement $C^{-1}(p - \tau_t)$ as the relevant state variable:

$$V(Y) = \max_{\tau \geq 0} E\left[\int_{C^{-1}(p-\tau_t)}^{Z_t G(C^{-1}(p-\tau_{t-1}))} \pi(s) ds + \beta V(Z_{t+1} G(C^{-1}(p - \tau_t))) \right] \tag{31}$$

Note that the profit reduction from the tax and the tax revenue net each other out. After this consideration, I take the first-order condition concerning τ_t:

$$\frac{\partial V(.)}{\partial \tau_t} = 0 = E\left[-\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] + \beta E\left[\frac{d}{d\tau_t}[V_{t+1}(.)]\right] = 0 \tag{32}$$

Then, taking the FOC concerning τ_{t-1}, and using Leibniz integral rule:

$$\frac{\partial V_t(.)}{\partial \tau_{t-1}} = E\left[\left(p - C(Z_t G(y_{t-1}(\tau_{t-1})))\right)\frac{d}{d\tau_{t-1}}\left(Z_t G(y_{t-1}(\tau_{t-1}))\right)\right] \tag{33}$$

By the Benveniste-Scheinkman theorem, I move forward the expression 7 by one period:

$$\frac{\partial V_{t+1}(.)}{\partial \tau_t} = E\left[\left(p - C(Z_{t+1} G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1} G(y_t(\tau_t))\right)\right]$$

Putting everything together:

$$\begin{aligned}
E\left[\left(p - C(y_t(\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
E\left[\left(p - C(C^{-1}(p-\tau_t))\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \text{ Replace } y_t(\tau_t) \\
E\left[\left(p - (p-\tau_t)\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
E\left[\left(\tau_t\right)\frac{d}{d\tau_t}\left(y_t(\tau_t)\right)\right] &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\frac{d}{d\tau_t}\left(Z_{t+1}G(y_t(\tau_t))\right)\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[\frac{dZ_{t+1}}{d\tau_t}G(y_t(\tau_t)) + Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\frac{dy_t}{d\tau_t}\right]\right] \\
\tau_t\frac{dy_t}{d\tau_t} &= \beta E\left[\left(p - C(Z_{t+1}G(y_t(\tau_t)))\right)\left[Z_{t+1}\frac{dG(y_t)}{dy_t}\right]\right]\frac{dy_t}{d\tau_t} \\
\tau_t &= \beta E\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t} - C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] \\
\tau_t &= \beta E\left[\left(pZ_{t+1}\frac{dG(y_t)}{dy_t}\right)\right] - \beta E\left[C(Z_{t+1}G(y_t(\tau_t)))Z_{t+1}\frac{dG(y_t)}{dy_t}\right] \\
\tau_t &= \beta p\frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1}C(Z_{t+1}G(y_t))\right]\frac{dG(y_t)}{dy_t}
\end{aligned}$$

In the derivation above, the expression is conditioned on period t information, and taxes do not influence shocks [10].

[10] A more stringent assumption is assuming independence between Z, G, and C

To make the result comparable with Reed (1979), I employ a cost function of the family $C(s) = \frac{\theta}{s^\alpha}$. Notice this function is homogeneous $C(ms) = \frac{\theta}{(ms)^\alpha} = m^{-\alpha}\frac{\theta}{(s)^\alpha}$. Then, the expression above can be expressed in two different ways according to catchability assumptions:

$$\begin{aligned}
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1} C(Z_{t+1} G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1} Z_{t+1}^{-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta p \frac{dG(y_t)}{dy_t} - \beta E\left[Z_{t+1}^{1-\alpha} C(G(y_t))\right] \frac{dG(y_t)}{dy_t} \\
\tau_t &= \beta \left[p - E\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t}
\end{aligned}$$

For the element $E\left[Z_{t+1}^{1-\alpha}\right]$, I have 2 options on how to proceed:

- Assume constant catchability $\alpha = 1$, then $E\left[Z_{t+1}^{1-1}\right] = 1$, then the tax does not depend on the current stocks: $\tau_t = \beta\left[p - (C \circ G)(y_t)\right] \frac{dG(y_t)}{dy_t}$.
- Assume decreasing catchability $0 < \alpha < 1$, and then apply Jensen's Inequality: $E\left[Z_{t+1}^{1-\alpha}\right] \leq E\left[Z_{t+1}\right]^{1-\alpha} = 1$. Hence, $\tau_t \geq \beta\left[p - (C \circ G)(y_t)\right] \frac{dG(y_t)}{dy_t}$ is a lower bound (a minimum tax).

Notice that the LHS is equivalent to part of the Euler equation for constant escapement proposed by Clark (1976). Then, it is possible to propose a constant tax based on the constant optimal escapement that satisfies the Euler equation for y^*:

$$\begin{aligned}
\tau_t &= \beta \left[p - E\left[Z_{t+1}^{1-\alpha}\right] C(G(y_t))\right] \frac{dG(y_t)}{dy_t} = (p - C(y_t)) \\
\tau^* &= \beta \left[p - E\left[Z_{t+1}^{1-\alpha}\right] C(G(y^*))\right] \frac{dG(y^*)}{dy^*} = (p - C(y^*)) \\
\tau^* &= \pi(y^*)
\end{aligned}$$

The optimal tax depends only on the optimal escapement, as Weitzman (2002) shows.

Appendix B: Derivation of the optimal marginal profit $m\pi$

B.1: Fisherman's problem

In each period t, a myopic fisherman maximizes profits through harvesting N stocks $i = \{1, 2 \ldots, n\}$ with independent life cycles, considering that the planner uses an instrument that sets a target $m\pi_t$:

$$\begin{aligned} \pi_t = \max_{\{h_t^i\}_{i=1}^N} & \left\{ \sum_{i=1}^{N} \int_{X_t^i - h_t^i}^{X_t^i} \left(p^i - m\pi_t - C^i(s^i) \right) ds^i \right\} \\ \text{s.t. } & h_t^i \geq 0 \ \forall i \end{aligned} \tag{34}$$

where $m\pi_t$ is the target marginal profit that can be obtained through the landing fee per fish unit or a shared quota, applied to all species. From the FOC concerning each harvest, I obtain the optimality condition for the current period:

$$p^1 - C^1\left(X_t^1 - h_t^1\right) = p^2 - C^2\left(X_t^2 - h_t^2\right) = \cdots = p^n - C^n\left(X_t^n - h_t^n\right) = m\pi_t$$

Using the definition of escapement $y_t^i = X_t^i - h_t^i$, I can re-express this condition as:

$$p^1 - C^1\left(y_t^1\right) = p^2 - C^2\left(y_t^2\right) = \cdots = p^n - C^n\left(y_t^n\right) = m\pi_t \tag{35}$$

Notice that Equation 35 is the equimarginal principle: the total profits are maximized when the marginal profit of the last harvested unit for each species i is equal. Like in the previous section, the fisher sets the escapement decision:

$$\begin{aligned} p^i - m\pi_t - C^i(y_t^i) &= 0 \\ C^i(y_t^i) &= p^i - m\pi_t \\ y_t^i &= (C^i)^{-1}(p^i - m\pi_t) \end{aligned} \tag{36}$$

The fisherman's escapement decision is independent of stock size and, therefore, unaffected by random ecological shocks.

B.2: Social planner's problem

The social planner incorporates the above fisherman's escapement decision into her dynamic optimization problem:

$$V_t(\vec{Y_t}) = \max_{\tau_t \geq 0} \ \mathbb{E}\left\{ \sum_{i=1}^{N} \int_{Y^i(m\pi_t)}^{Z_t^i G^i(Y_{t-1}^i(m\pi_{t-1}))} \left(p^i - C^i(s^i) \right) ds^i + \beta V_{t+1}(\vec{X}_{t+1}) \right\} \tag{37}$$

The target marginal profit depends on the available information. If the manager does not know the current stock status, I assume they operate based on each y^i_{t-1}. Although the planner cannot observe current ecological shocks, I include them in the dynamic problem to evaluate their impact on the tax basket. I take the FOC with respect to $m\pi_t$ (following the same procedure as in Appendix A) and obtain the following Euler equation:

$$\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y^i_t))y^i_{m\pi_t}\Big] = \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big]$$

Replacing the fisherman's condition $y^i_t = (C^i)^{-1}(p^i - m\pi_t)$:

$$\begin{aligned}
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(y^i_t))y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i((C^i)^{-1}(p^i - m\pi_t)))y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big] \\
\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - (p^i - m\pi_t))y^i_{m\pi_t}\Big] &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big] \\
\sum_{i=1}^{N} m\pi_t y^i_{m\pi_t} &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big] \\
m\pi_t \sum_{i=1}^{N} y^i_{m\pi_t} &= \beta\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big] \\
m\pi^+_t &= \beta\frac{1}{\sum_{j=1}^{N} y^j_{m\pi_t}}\mathbb{E}\Big[\sum_{i=1}^{N}(p^i - C^i(Z^i_t G^i(y^i_t)))Z^i_t G^i_{y^i_t} y^i_{m\pi_t}\Big] \\
m\pi^+_t &= \sum_{i=1}^{N} m\pi_{t+1}\frac{y^i_{m\pi_t}}{\sum_{j=1}^{N} y^j_{m\pi_t}}
\end{aligned} \tag{38}$$

If the cost function is $C^i(s^i_t) = \frac{\theta^i}{s^i_t}$, then $y^i_{m\pi_t} = \frac{\theta^i}{(p^i - m\pi)^2}$

$$\begin{aligned}
m\pi^+_t &= \sum_{i=1}^{N} m\pi_{t+1}\frac{y^i_{m\pi_t}}{\sum_{j=1}^{N} y^j_{m\pi_t}} \\
&= \sum_{i=1}^{N} m\pi_{t+1}\frac{\frac{\theta^i}{(p^i - m\pi)^2}}{\sum_{j=1}^{N}\frac{\theta^j}{(p^j - m\pi)^2}}
\end{aligned} \tag{39}$$

I know the fisherman decision is $p^i - m\pi = C^i(y^i)$ and $C^i(y^i) = \frac{\theta^i}{y^i}$:

$$\begin{aligned} m\pi_t^+ &= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\theta^i}{(p^i - m\pi)^2}}{\sum_{j=1}^{N} \frac{\theta^j}{(p^j - m\pi)^2}} \\ &= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\theta^i}{C^i(y^i)^2}}{\sum_{j=1}^{N} \frac{\theta^j}{C^j(y^j)^2}} \\ &= \sum_{i=1}^{N} m\pi_{t+1} \frac{\frac{\left(Y^i(m\pi^+)\right)^2}{\theta^i}}{\sum_{j=1}^{N} \frac{\left(Y^j(m\pi^+)\right)^2}{\theta^j}} \\ &= \sum_{i=1}^{N} m\pi_{t+1} \ \rho_t^i \end{aligned} \tag{40}$$

The weights per species depend on its respective unfished biomass and cost parameter.

B.3: Tax basket equivalence

Replacing $m\pi = \tau_t$, it is straightforward to obtain the same expression as in Appendix B.2:

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p - E\left[Z_{t+1}^{i,1-\alpha} \right] C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}$$

For the element $E\left[Z_{t+1}^{1-\alpha}\right]$, I have 2 options on how to proceed:

- Assume constant catchability $\alpha = 1$, then $E\left[Z_{t+1}^{1-1}\right] = 1$, so the tax does not depend on the current stocks.
- Assume decreasing catchability $0 < \alpha < 1$, and then apply Jensen's Inequality: $E\left[Z_{t+1}^{1-\alpha}\right] \leq E\left[Z_{t+1}\right]^{1-\alpha} = 1$. Hence, τ_t is a lower bound (a minimum tax).

The tax basket is the weighted average of the marginal profits per species in the next period. For simplicity, assuming $\alpha = 1$ and a cost function $C^i = \frac{\theta^i}{s^i}$ provides:

- $y_t^i = \frac{\theta^i}{p^i - \tau_t}$
- $\frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$

However, from the firm's optimization, I know that $p^i - \tau_t = C(y_t^i)$ and $C(y_t^i) = \frac{\theta^i}{y_t^i}$. With this information, I can rearrange the tax basket:

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p - E\left[Z_{t+1}^{i,1-\alpha} \right] C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{dy_t^i}{d\tau_t}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{dy_t^i}{d\tau_t} \qquad \text{Replace } \alpha = 1$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{(p^i - \tau_t)^2}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{\theta^i}{(p^i - \tau_t)^2} \qquad \text{Replace } \frac{dy_t^i}{d\tau_t} = \frac{\theta^i}{(p^i - \tau_t)^2}$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{\theta^i}{C^i(y_t^i)^2}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{\theta^i}{C^i(y_t^i)^2} \qquad \text{Replace } p^i - \tau_t = C^i(y_t^i)$$

$$\tau_t = \beta \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i)^2}{\theta^i}} \sum_{i=1}^{N} \left[p - C(G(y_t^i)) \right] \frac{dG(y_t^i)}{dy_t^i} \frac{(y_t^i)^2}{\theta^i} \qquad \text{Replace } C^i(y_t^i) = \frac{\theta^i}{y_t^i}$$

$$\tau_t = \frac{1}{\sum_{i=1}^{N} \frac{(y_t^i(\tau_t))^2}{\theta^i}} \sum_{i=1}^{N} \pi_{t+1}^i(y_t^i(\tau_t)) \frac{(y_t^i(\tau_t))^2}{\theta^i} \qquad \text{Rename next period's marginal profit}$$

$$\tau_t = \tau_t(\vec{y_t^i})$$

Similar to $m\pi_t$, after employing Proposition 1, the tax basket is also a function of each species' escapement. If the species are identical, it is straightforward to verify that the tax basket equals the landing fee for individual species, as shown in Appendix A. Moreover, the social planner need not know the ecological shocks in the current period.

Appendix C: Auxiliary proofs

Appendix C.1: Welfare rearrangement

By Lemma 2.1 the fisherman equates $p^i - C^i(y_t^i) = \tau_t = \lambda_t$, so $y_t^i = m\pi_t = Y^i(\tau_t) = Y^i(\lambda_t)$. If $X_t^i > y_t^i$ for all i, this Lemma holds every period. Therefore, escapement depends on the current tax, not on the ecological shocks. With $C^i(s) = \theta^i/s$, the fisherman's myopic payoff is

$$\int_{y_t^i}^{X_t^i} \Big(p^i - \frac{\theta^i}{s}\Big) ds = A^i(X_t^i) + \phi^i(y_t^i), \quad \text{where } A^i(X) := p^i X_t^i - \theta^i \ln X_t^i, \quad \phi^i(y_t^i) := -p^i y + \theta^i \ln y_t^i,$$

which is 14. The current stock and the current escapement enter additively. Substituting $X_t^i = Z_t^i G^i(y_{t-1}^i)$, using $\mathbb{E}[Z_t^i] = 1$, $\mathbb{E}|\ln Z_t^i| < \infty$. and $\ln(Z_t^i G^i) = \ln Z_t^i + \ln G^i$,

$$\begin{aligned}
\mathbb{E}\big[A^i\big(Z_t^i G^i(y_{t-1}^i)\big)\big] &= \mathbb{E}\big[p^i X - \theta^i \ln X\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln(Z_t^i G^i)\big] \\
&= \mathbb{E}\big[p^i Z_t^i G^i - \theta^i \ln G^i - \theta^i \ln Z_t^i\big] \\
&= p^i G^i(y_{t-1}^i) - \theta^i \ln G^i(y_{t-1}^i) - \theta^i \,\mathbb{E}[\ln Z_t^i] \\
&= a^i(y_{t-1}^i) - \kappa^i
\end{aligned}$$

with $a^i(y) := p^i G^i(y) - \theta^i \ln G^i(y)$, $\kappa^i := \theta^i \mathbb{E}[\ln Z_t^i]$ and $\mathbb{E}[\ln Z_t^i] < \infty$ by assumption. Then, taking the initial stock X_0^i as given (and y_0^i as given) and summing across all species i,

$$\mathcal{W} = \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] \Big\}$$

Shifting the index in $\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i) = \beta \sum_{t\geq 0} \beta^t a^i(y_t^i)$ and reallocating β^t,

$$\begin{aligned}
\sum_{t\geq 1} \beta^t \big[a^i(y_{t-1}^i) - \kappa^i + \phi^i(y_t^i)\big] &= \underbrace{\sum_{t\geq 1} \beta^t a^i(y_{t-1}^i)}_{\text{(I): stock block}} - \underbrace{\sum_{t\geq 1} \beta^t \kappa^i}_{\text{(II): shock block}} + \underbrace{\sum_{t\geq 1} \beta^t \phi^i(y_t^i)}_{\text{(III): escapement block}} \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \sum_{t\geq 1} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \beta \sum_{t\geq 0} \beta^t + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i)
\end{aligned}$$

Putting things together:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) + \phi^i(y_0^i) + \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \Big\} \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_i \beta \sum_{t\geq 0} \beta^t a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_i \sum_{t\geq 1} \beta^t \phi^i(y_t^i) \\
&= \sum_i A^i(X_0^i) + \sum_i \phi^i(y_0^i) + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) - \sum_i \kappa^i \frac{\beta}{1-\beta} + \sum_{t\geq 1} \beta^t \sum_i \phi^i(y_t^i) \\
&= \sum_i [A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}] + \sum_{t\geq 0} \beta^t \sum_i \beta a^i(y_t^i) + \sum_{t\geq 0} \beta^t \sum_i \phi^i(y_t^i) \\
&= \sum_i [\underbrace{A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta}}_{=:\Xi_0}] + \sum_{t\geq 0} \beta^t \sum_i [\underbrace{\phi^i(y_t^i) + \beta a^i(y_t^i)}_{=:w^i(y_t^i)}] \\
&= \Xi_0 + \sum_{t\geq 0} \beta^t \sum_i w^i(y_t^i)
\end{aligned}
\tag{41}
$$

This result applies to a tax or quota basket.

Appendix C.2: Recovering the dynamic solution through the static problem

Reopening equation 41 and assuming that $w^i(y_t^i)$ is constant across time:

$$
\begin{aligned}
\mathcal{W} &= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[\phi^i(y^i) + \beta a^i(y^i) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \frac{1}{1-\beta} \Big[\beta \big(p^i G^i(y^i) - \theta^i \ln G^i(y^i) \big) - \big(p^i y^i - \theta^i \ln y^i \big) \Big] \Big\} \\
&= \sum_i \Big\{ A^i(X_0^i) - \kappa^i \frac{\beta}{1-\beta} + \underbrace{\frac{1}{1-\beta} \Big[\beta p^i G^i(y^i) - \beta \theta^i \ln G^i(y^i) - p^i y^i + \theta^i \ln y^i \Big]}_{\mathcal{V}^i} \Big\}
\end{aligned}
\tag{42}
$$

I take the derivative with respect to y:

$$
\begin{aligned}
\frac{d\mathcal{W}}{dy} &= \sum_i \frac{1}{1-\beta}\Big[\beta p^i G^{i\prime}(y^i) - \beta\theta^i \frac{1}{G^i(y^i)} G^{i\prime}(y^i) - p^i + \theta^i \frac{1}{y^i}\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\beta[p^i G^{i\prime}(y^i) - \theta^i \frac{1}{G^i(y^i)} G^{i\prime}(y^i)] - p^i + \theta^i \frac{1}{y^i}\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\beta[p^i - \theta^i \frac{1}{G^i(y^i)}] G^{i\prime}(y^i) - [p^i - \theta^i \frac{1}{y^i}]\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\beta[p^i - C^i(G^i(y^i))] G^{i\prime}(y^i) - [p^i - C^i(y^i)]\Big] \\
&= \sum_i \frac{1}{1-\beta}\Big[\pi^i_{t+1}(y^i) - [p^i - C^i(y^i)]\Big]
\end{aligned}
\tag{43}
$$

I recover the Euler equation of Proposition 2. Moreover, ignoring the summation notation to focus on the species-specific management, the discount factor multiplies the growth block in 42; this is what makes 43 vanish exactly at Reed's y^{i*}. Because $\pi^{i\prime} \leq 0$ and $C^{i\prime} < 0$, so $\mathcal{V}^{i\prime\prime} < 0$: $\mathcal{V}^i$ is strictly concave with unique maximizer y^{i*}, then,

$$
\pi^i_{t+1}(y^i) > p^i - C^i(y^i) \;\text{ for } y^i < y^{i*}, \qquad \pi^i_{t+1}(y^i) < p^i - C^i(y^i) \;\text{ for } y^i > y^{i*} \tag{44}
$$

This component is key to proving uniqueness.

Appendix D: Existence, uniqueness and interiority of τ^+

By Proposition 2 and Appendix C.2, the regularity condition $\max_i \tau^{i*} < \tau^F_{\max} := \min_i(p^i - C^i(K^i))$, and the cost function $C^i = \theta^i/s^i$, I can reinterpret the tax optimality condition as a residual function:

$$g(\tau) := \tau - R(\tau), \qquad \text{where: } R(\tau) := \frac{\sum_i \pi^i(Y^i(\tau))\,(Y^i(\tau))^2/\theta^i}{\sum_i (Y^i(\tau))^2/\theta^i},$$

so $g(\tau)$ and $\mathcal{W}'(\tau)$ share sign and roots.

Appendix D.1: Endpoint signs

From Appendix C.2, I know the tax solution commits species i to a constant escapement y forever. Without loss of generality, order species according to their individual optimal taxes $\tau^{1*} \leq \cdots \leq \tau^{N*}$ and assume not all τ^{i*} are equal. Then, I can establish the following logical sequence:

1. $Y^i(\cdot)$ is strictly increasing, so $\tau < \tau^{i*} \iff Y^i(\tau) < y^{i*}$ (lower tax, less remaining stock in the ocean).

2. At $\tau = \tau^{1*}$: species 1 contributes exactly $\pi^1_{t+1}(Y^1(\tau^{1*})) = \tau^{1*}$ (shown in Appendix A).

3. Each $j \geq 2$ has $\tau^{1*} < \tau^{j*}$, hence $Y^j(\tau^{1*}) < y^{j*}$ (over-harvest), and 44 with $p^j - C^j(Y^j(\tau^{1*})) = \tau^{1*}$ gives $\pi^j_{t+1}(Y^j(\tau^{1*})) > \tau^{1*}$.

4. A weighted average with strictly positive weights (thanks to the fishable range) that are all $\geq \tau^{1*}$, at least one strictly, exceeds τ^{1*}. Therefore $R(\tau^{1*}) > \tau^{1*}$ and $g(\tau^{1*}) < 0$.

5. Starting from the opposite endpoing, $\tau = \tau^{N*}$: symmetrically, each $j < N$ has $Y^j(\tau^{N*}) > y^{j*}$ (under-harvest) so $\pi^j_{t+1}(Y^j(\tau^{N*})) < \tau^{N*}$, giving $R(\tau^{N*}) < \tau^{N*}$ and $g(\tau^{N*}) > 0$.

Appendix D.2: Existence

Y^i, π^i and the denominator $\sum_i \theta^i/(p^i - \tau)^2$ are continuous on $[\tau^{1*}, \tau^{N*}]$. The interior regularity guarantees $p^i - \tau \geq C^i(K^i) > 0$; then the denominator is bounded and bounded away from zero, and g is continuous. By the Intermediate Value Theorem there exists $\tau^+ \in (\tau^{1*}, \tau^{N*})$ with $g(\tau^+) = 0$.

Appendix D.3: Uniqueness

The sufficient condition is that $\mathcal{W}$ be strictly concave on $[\tau^{1*}, \tau^{N*}]$ as shown in Appendix C.2. However, concavity of each w^i in its respective escapement is not enough, because τ enters w^i through Y^i, and $Y^{i\prime}$ sets the weights. Without loss of generality, I define and focus on $W(\tau) = \sum_i w^i(Y^i(\tau))$. Differentiating W twice:

$$W''(\tau) = \sum_i \Big[\underbrace{w^{i\prime\prime}(y^i)\left(Y^{i\prime}(\tau)\right)^2}_{\leq 0 \text{ if } w^i \text{ concave}} + \underbrace{w^{i\prime}(y^i)\, Y^{i\prime\prime}(\tau)}_{\text{indefinite}} \Big], \qquad y^i = Y^i(\tau) \tag{45}$$

Notice that y^i differ across species even though τ is shared, and the sum contains no cross term (W is separable). Then, the global concavity is equivalent to per-species concavity. However, the term $w^{i\prime}(y^i)\, Y^{i\prime\prime}(\tau)$ has ambiguous sign: $Y^{i\prime\prime}(\tau) = 2\theta^i/(p^i - \tau)^3 > 0$, while $w^{i\prime}(y^i) = \pi^i_{t+1}(y^i) - \tau$ is positive below τ^{i*} and negative above it by 44.

To treat this, I remove the nonlinearity of Y^i through affine reparametrization. From the fisherman's rule $p^i - \theta^i/y^i = \tau$,

$$p^i - \frac{\theta^i}{y^i} = \tau$$
$$z^i := \frac{1}{y^i} = \frac{p^i - \tau}{\theta^i}$$

is affine in τ, so $z^{i\prime\prime}(\tau) = 0$ and the curvature term vanishes. With this, I rewrite W:

$$W(\tau) = \sum_i w^i(y^i) = \sum_i w^i(1/z^i) = \sum_i \hat{w}^i(z^i), \quad \text{where } \hat{w}^i(z) := w^i(1/z).$$

Because each z^i is affine in τ and W is separable, $W''(\tau) = \sum_i \hat{w}^{i\prime\prime}(z^i)\,(z^{i\prime})^2$ with $(z^{i\prime})^2 > 0$. Then, W is strictly concave in $\tau \iff \hat{w}^{i\prime\prime}(z^i) \leq 0$ for all i. Using $\dfrac{dy}{dz} = \dfrac{d}{dz} z^{-1} = -\dfrac{1}{z^2}$ and $z = 1/y \Rightarrow \dfrac{1}{z^2} = y^2$, I get,

$$\hat{w}^{i\prime}(z) = w^{i\prime}\left(\tfrac{1}{z}\right)\left(-\tfrac{1}{z^2}\right) = -\tfrac{1}{z^2}\, w^{i\prime}(y)$$

and

$$\hat{w}^{i\prime\prime}(z) = \tfrac{2}{z^3} w^{i\prime}\left(\tfrac{1}{z}\right) - \tfrac{1}{z^2} w^{i\prime\prime}\left(\tfrac{1}{z}\right)\left(-\tfrac{1}{z^2}\right) = \tfrac{2}{z^3} w^{i\prime}\left(\tfrac{1}{z}\right) + \tfrac{1}{z^4} w^{i\prime\prime}\left(\tfrac{1}{z}\right).$$

Replacing $z = 1/y$ (so $1/z^3 = y^3$, $1/z^4 = y^4$),

$$\begin{aligned} \hat{w}^{i\prime\prime}(z) &= 2y^3 w^{i\prime}(y) + y^4 w^{i\prime\prime}(y) \\ &= y^3 \left[\, y\, w^{i\prime\prime}(y) + 2 w^{i\prime}(y) \,\right] \end{aligned}$$

From Appendix C.2, I know $w^{i\prime}(y) = \pi^i_{t+1}(y) - p^i + \theta^i/y$ and $w^{i\prime\prime}(y) = \pi^{i\prime}_{t+1}(y) - \theta^i/y^2$,

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= y^3\big[\,y\,(\pi^{i\prime}_{t+1}(y) - \theta^i/y^2) + 2(\pi^i_{t+1}(y) - p^i + \theta^i/y)\big] \\ &= y^3\big[y\pi^{i\prime}_{t+1}(y) - \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i + 2\theta^i/y\big] \\ &= y^3\big[y\pi^{i\prime}_{t+1}(y) + \theta^i/y + 2\pi^i_{t+1}(y) - 2p^i\big] \\ &= y^3\big[y\pi^{i\prime}_{t+1}(y) + C^i(y) + 2\pi^i_{t+1}(y) - 2p^i\big].\end{aligned}$$

Using the fisherman's species-specific optimality condition $C^i(Y^i(\tau)) = p^i - \tau$,

$$\begin{aligned}\hat{w}^{i\prime\prime}(z) &= y^3\big[y\pi^{i\prime}_{t+1}(y) + p^i - \tau + 2\pi^i_{t+1}(y) - 2p^i\big] \\ &= \underbrace{y^3}_{>0}\,\underbrace{\big[y\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) - p^i - \tau\big]}_{\text{sign of } \hat{w}^{i\prime\prime}}.\end{aligned}$$

Since $y^3 > 0$, $\hat{w}^{i\prime\prime}(z) \le 0$ if and only if the bracket is ≤ 0, which is:

Condition for global concavity For every i and every $\tau \in [\tau^{1*}, \tau^{N*}]$, writing $y = Y^i(\tau)$,

$$y\,\pi^{i\prime}_{t+1}(y) + 2\pi^i_{t+1}(y) \;\le\; p^i + \tau, \tag{46}$$

with strict inequality for at least one i. Because W is separable, imposing (46) for every i makes $W''(\tau) \le 0$.

Appendix D.4: Interiority

The endpoint signs $g(\tau^{1*}) < 0 < g(\tau^{N*})$ are strict, so $\tau^+ \in (\tau^{1*}, \tau^{N*}) \subset (0, \tau^F_{\max})$. Then, the optimum is never at $\min_i \tau^{i*}$ or $\max_i \tau^{i*}$ unless these coincide.

Appendix E: Derivation of the optimal quota basket

E.1: Fisherman's problem

Each period, the fisherman maximizes profits over the harvest of N stocks subject to the shared quota:

$$\begin{aligned} &\max_{h^i} \left\{ \pi_t = \sum_i p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s_t^i)\, ds^i \right\} \\ &\text{subject to} \quad \omega_t = \sum_i h_t^i = \sum^2 X_t^i - \sum^2 y_t^i \end{aligned} \tag{47}$$

The Lagrangian is:

$$\max_{h^i} \left\{ \sum_{i=1} p^i h_t^i - \int_{y_t^i}^{X_t^i} C^i(s_t^i)\, ds^i - \lambda_t \Big(\omega_t - \sum h_t^i \Big) \right\},$$

with FOC for each stock

$$\begin{aligned} p^i - C^i(y_t^i) &= \lambda_t \\ y_t^i &= (C^i)^{-1}(p^i - \lambda_t) = Y^i(\lambda_t) \end{aligned} \tag{48}$$

The realized stocks $X_t^i = Z_t^i G^i(y_{t-1}^i)$ depend on last period's escapement and this period's shock, not on the current quota:

$$\begin{aligned} \omega_t &= \sum_i X_t^i - \sum_i Y^i(\lambda_t) \\ \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \lambda_t &= \lambda(\mathbf{X}_t, \omega_t) \end{aligned} \tag{49}$$

Differentiating (49) with respect to ω_t, holding the predetermined stocks $\mathbf{X}_t$ fixed (because the fisherman observes them),

$$\begin{aligned} \sum_i Y^i(\lambda_t) &= \sum_i X_t^i - \omega_t \\ \frac{\partial}{\partial \omega} \Big(\sum_i Y^i(\lambda_t) \Big) &= \frac{\partial}{\partial \omega} \Big(\sum_i X_t^i - \omega_t \Big) \\ \Big(\sum_i Y_\lambda^i \Big) \lambda_{\omega_t} &= -1 \\ \lambda_{\omega_t} &= \frac{-1}{\sum_i Y_\lambda^i} < 0 \end{aligned} \tag{50}$$

If the cost function is $C^i = \theta^i/s^i$, $Y^i_\lambda = \theta^i/(p^i-\lambda)^2 = (y^i)^2/\theta^i > 0$ because $p^i - \lambda = C^i(y^i) = \theta^i/y^i$ and $y^i, \theta^i > 0$ by definition. Consequently, a larger quota lowers the common shadow value. A change in the harvest limit causes the following change in the species-specific escapement:

$$\begin{aligned} y^i_t &= Y^i(\lambda_t) \\ \frac{\partial y^i_t}{\partial \omega_t} &= Y^i_\lambda \, \lambda_{\omega_t} \\ &= Y^i_\lambda \; \frac{-1}{\sum_j Y^j_\lambda} \qquad \text{Replace (50)} \\ &= -\frac{(y^i_t)^2/\theta^i}{\sum_j (y^j_t)^2/\theta^j} \\ &= -\rho^i_t \qquad \text{Rename} \end{aligned} \tag{51}$$

Notice that $\sum_i \rho^i_t = 1$ if the regularity condition holds. The weights ρ^i_t record how the fisherman spreads one extra unit of quota basket across species.

E.2: Choosing the quota is choosing the shared shadow value

The total profits per period are:

$$\Pi(\mathbf{X}_t, \omega_t) := \sum_i \int_{y^i_t}^{X^i_t} \left(p^i - C^i(s)\right) ds$$

with $y^i_t = Y^i(\lambda_t)$. Deriving with respect to ω, using the fisherman's FOC $p^i - C^i(y^i_t) = \lambda_t$ and $\sum_i \partial y^i_t/\partial \omega_t = -1$ from (49),

$$\begin{aligned} \frac{\partial \Pi}{\partial \omega_t} &= -\sum_i \left(p^i - C^i(y^i_t)\right) \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\sum_i \lambda_t \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\lambda_t \sum_i \frac{\partial y^i_t}{\partial \omega_t} \\ &= -\lambda_t \sum_i (-\rho^i_t) \\ &= \lambda_t \end{aligned} \tag{52}$$

The marginal social value of an extra unit of aggregate quota equals the fisherman's shadow value λ_t. This is the duality between the two derivations: λ_t is the price that decentralizes the quota.

E.3: Social Planner's problem

Take the predetermined escapement vector $\vec{y}_{t-1}$ as the state. The planner sets the quota ω_t before the shocks, so ω_t is deterministic while λ_t and y_t^i are random through $\mathbf{X}_t$:

$$V(\vec{y}_{t-1}) = \max_{\omega_t \geq 0} \mathbb{E}\Big[\Pi(\mathbf{X}_t, \omega_t) + \beta V(\vec{y}_t)\Big], \qquad X_t^i = Z_t^i G^i(y_{t-1}^i), \quad y_t^i = Y^i\big(\lambda(\mathbf{X}_t, \omega_t)\big) \tag{53}$$

Only y_t^i depends on ω_t (the stocks $\mathbf{X}_t$ are predetermined on y_{t-1}^i). Using $\partial\Pi/\partial\omega_t = \lambda_t$ from 52 and $\partial y_t^i/\partial\omega_t = -\rho_t^i$ from 51, I derive the FOC concerning the quota basket,

$$\begin{aligned} \frac{\partial V(\cdot)}{\partial \omega_t} &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta[V_{y^1} y_\lambda^1 \lambda_\omega + V_{y^2} y_\lambda^2 \lambda_\omega + ... + V_{y^n} y_\lambda^n \lambda_\omega]\Big] = 0 \\ &= \mathbb{E}\Big[\frac{\partial \Pi}{\partial \omega_t} + \beta \sum_j V_{y^j}(\vec{y}_t)\, \frac{\partial y_t^j}{\partial \omega_t}\Big] = 0 \\ &= \mathbb{E}\Big[\lambda_t - \beta \sum_j \rho_t^j\, V_{y^j}(\vec{y}_t)\Big] = 0 \end{aligned} \tag{54}$$

The optimal quota basket equates, in expectation, the current shadow value of the quota with the ρ-weighted continuation value of the escapement.

The state y_{t-1}^k enters 53 through $X_t^k = Z_t^k G^k(y_{t-1}^k)$. Differentiating 49 with respect to the stock, holding ω_t fixed,

$$\Big(\sum_j Y_\lambda^j\Big)\frac{\partial \lambda_t}{\partial X_t^k} = 1, \qquad \frac{\partial y_t^j}{\partial X_t^k} = Y_\lambda^j \frac{\partial \lambda_t}{\partial X_t^k} = \frac{Y_\lambda^j}{\sum_m Y_\lambda^m} = \rho_t^j, \qquad \sum_j \frac{\partial y_t^j}{\partial X_t^k} = 1 \tag{55}$$

Applying the envelope theorem (the indirect effect through the optimal ω_t is zero by 54) and the chain rule,

$$
\begin{aligned}
V_{y^k_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(\frac{\partial \Pi}{\partial X^k_t} + \beta \frac{\partial V(\vec{y}_t)}{\partial X^k_t}\Big) \underbrace{Z^k_t G^{k\prime}(y^k_{t-1})}_{\partial X^k_t / \partial y^k_{t-1}}\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\frac{\partial X^i_t}{\partial X^k_t} - \sum_i \big(p^i - C^i(y^j_t)\big)\frac{\partial y^j_t}{\partial \lambda_t}\frac{\partial \lambda_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^i_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)\delta_{ik} - \sum_i \big(p^i - C^i(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^i_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(\sum_i \big(p^i - C^i(X^i_t)\big)1 - \sum_i \big(p^i - C^i(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^i_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \sum_j \big(p^j - C^j(y^j_t)\big)\frac{\partial y^j_t}{\partial X^k_t} \;+\; \beta \sum_j V_{y^j}(\vec{y}_t)\frac{\partial y^j_t}{\partial X^k_t}\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t \sum_j \rho^j_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X^k_t) - \lambda_t + \beta \sum_j \rho^j_t V_{y^j}(\vec{y}_t)\Big) Z^k_t G^{k\prime}(y^k_{t-1})\Big]
\end{aligned}
\tag{56}
$$

using $\partial X^i_t / \partial X^k_t = \delta_{ik}$ for the stocks (only stock k responds to its respective y^k_{t-1}) and $\partial y^i_t / \partial X^k_t = \rho^i_t$ from 55 for the quota-coupled escapements (all respond, through λ_t). The bracket is the shadow value of one extra fish in stock k minus the λ_t of the escapement it displaces, plus its continuation value. The escapements are all tied through the quota basket.

Equations 54–56 are the direct-ω analog to Appendix 4.

E.4: Recovering λ^+

With $Z^i_t \equiv 1$ the expectation drops and 54 holds pointwise, $\lambda_t = \beta \sum_i \rho^i_t V_{y^i}(\vec{y}_t)$. Substituting this into 56 cancels the last two bracketed terms, leaving

$$
\begin{aligned}
V_{y^i_{t-1}}(\vec{y}_{t-1}) &= \mathbb{E}_{Z_t}\Big[[p^i - C^i(X^i_t) - \lambda_t + \beta \sum_i \rho^i_t V_{y^i}(\vec{y}_t)] Z^i_t G^{i\prime}(y^i_{t-1})\Big] \\
&= \Big[[p^i - C^i(X^i_t) - \lambda_t + \lambda_t] G^{i\prime}(y^i_{t-1})\Big] \\
&= [p^i - C^i(X^i_t)] G^{i\prime}(y^i_{t-1}) \\
&= \frac{1}{\beta}\pi^i(y^i_{t-1}) \quad \text{Using definition of } \pi^i = [p^i - C^i]G^{i\prime}
\end{aligned}
\tag{57}
$$

Substituting back into the FOC and using Remark 1.2,

$$
\begin{aligned}
\lambda_t &= \beta \sum_i \rho_t^i V_{y^i}(\vec{y}_t) \\
\lambda_t &= \sum_i \rho_t^i \, \pi^i(y_t^i) \\
\lambda^+ &= \frac{\sum_i \pi^i\big(y_t^i\big) \, (y_t^i)^2/\theta^i}{\sum_j (y_t^j)^2/\theta^j}
\end{aligned}
\tag{58}
$$

which is exactly the planner's FOC for the tax 13. Like the optimal tax basket, the optimal shadow value is fixed over time $\lambda_t = \lambda^+$ with $y_t^i = Y^i(\lambda^+)$ if the regularity condition holds.

E.5: Quota basket is a second best

Under uncertainty, the collapse of Appendix E.4 fails because the shadow value λ_t^F and the reallocation weights ρ_t^i are both functions of the realized shocks. The planner, by contrast, commits to a deterministic ω_t. The FOC residual is:

$$
R_t := \lambda_t^F - \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t),
$$

so that 54 is $\mathbb{E}[R_t] = 0$. Substituting into 56 and separating,

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - \lambda_t + \beta \sum_j \rho_t^j V_{y^j}(\vec{y}_t)\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\Big[\Big(p^k - C^k(X_t^k) - R_t\Big) Z_t^k G^{k\prime}(y_{t-1}^k)\Big] \\
&= \mathbb{E}\big[(p^k - C^k(X_t^k)) Z_t^k\big] G^{k\prime}(y_{t-1}^k) - \mathbb{E}\big[R_t Z_t^k\big] G^{k\prime}(y_{t-1}^k)
\end{aligned}
$$

With $C^k(s) = \theta^k/s$ and $X_t^k = Z_t^k G^k(y_{t-1}^k)$ the shock cancels in the first expectation,

$$
\mathbb{E}\Big[\Big(p^k - \frac{\theta^k}{Z_t^k G^k(y_{t-1}^k)}\Big) Z_t^k\Big] = p^k - \frac{\theta^k}{G^k(y_{t-1}^k)} = p^k - C^k\big(G^k(y_{t-1}^k)\big)
$$

However, the second expression is a covariance

$$
\begin{aligned}
V_{y^k}(\vec{y}_{t-1}) &= \frac{1}{\beta} \pi_{t+1}^k(y_{t-1}^k) - \mathrm{Cov}\big(R_t, \, Z_t^k\big) \, G^{k\prime}(y_{t-1}^k) \\
\beta V_{y^k}(\vec{y}_{t-1}) &= \pi_{t+1}^k(y_{t-1}^k) - \beta \mathrm{Cov}\big(R_t, \, Z_t^k\big) \, G^{k\prime}(y_{t-1}^k)
\end{aligned}
\tag{59}
$$

In a deterministic setting, $\mathrm{Cov}(R_t, Z_t^k) = 0$ and 59 reduces to $\beta V_{y^k} = \pi^k$, which is Appendix E.4. Under uncertainty, the covariance is nonzero, because R_t moves with the shocks through both λ_t^F and ρ_t^j. The planner's single scalar ω_t can set the mean of R_t

to zero, but cannot remove its correlation with the shock vector. Given the deterministic quota,

$$\sum_i Y^i(\lambda_t^F) = \sum_i Z_t^i G^i(y_{t-1}^i) - \omega_t$$

So λ_t^F inherits the randomness of the aggregate realized stock. The quota choice

$$\omega_t = \sum_i G^i(y_{t-1}^i) - \sum_i Y^i(\lambda^+) \tag{60}$$

uses only $\mathbb{E}[Z_t^i] = 1$ and the predetermined y_{t-1}^i. It centers the distribution of λ_t^F near λ^+ and the escapements near their targets $\bar{y}^i = Y^i(\lambda^+)$, and coincides with the ω_t.

Appendix F: the bounds of the welfare loss

Lemma 4.1 (The welfare loss is bounded)**.** *$w^i(y) = \phi^i(y) + \beta a^i(y)$ is the dynamic per-species payoff of (41), with $\phi^i(y) = -p^i y + \theta^i \ln y$ and $a^i(y) = p^i G^i(y) - \theta^i \ln G^i(y)$. By (22), $w^{i\prime}(y) = \pi^i_{t+1}(y) - (p^i - C^i(y))$, so $w^{i\prime}(\bar{y}^i) = 0$, and $w^{i\prime\prime}(y) < 0$ on the fishable range $[\underline{y}^i, \hat{y}^i]$, where the bounds are the smallest and largest escapements the quota can induce (see Lemma 4.2), so that $0 < \underline{y}^i \leq \bar{y}^i \leq \hat{y}^i < \infty$. Then, for every $y \in [\underline{y}^i, \hat{y}^i]$:*

(a) The loss relative to the basket optimum is non-negative:

$$0 \;\leq\; w^i(\bar{y}^i) - w^i(y). \tag{61}$$

(b) The loss is at most a constant times the squared deviation:

$$w^i(\bar{y}^i) - w^i(y) \;\leq\; \tilde{B}^i\,(y - \bar{y}^i)^2, \qquad \tilde{B}^i = \tfrac{1}{2}\max_{[\underline{y}^i, \hat{y}^i]}\left(-w^{i\prime\prime}\right) < \infty. \tag{62}$$

The global maximizer of the concave $\bar{y}^i$ verifies (a). For part (b), I know:

$$w^i(b) - w^i(y) = \int_y^b w^{i\prime}(s)ds$$

renaming $u = w^{i\prime}(s)$ and $v = s - b$, then by integration by parts:

$$\begin{aligned}
\int_y^b w^{i\prime}(s)ds &= \Big[w^{i\prime}(s)(s-b)\Big]\Big|_b^y - \int_y^b (s-b)w^{i\prime\prime}(s)ds \\
&= -w^{i\prime}(y)(y-b) - \int_y^b (s-b)w^{i\prime\prime}(s)ds \\
&= w^{i\prime}(y)(b-y) - \int_y^b (s-b)w^{i\prime\prime}(s)ds
\end{aligned}$$

if $b = \bar{y}^i$, then $w^{i\prime} = 0$, then

$$w^i(\bar{y}^i) - w^i(y) = \int_y^{\bar{y}^i} \left(-w^{i\prime\prime}(s)\right)(s-y)\,ds,$$

The integrand is non-negative because $w^{i\prime\prime} < 0$. I control the curvature of $w^{i\prime\prime}$ with the maximum possible value in the range:

$$M^i := \max_{[\underline{y}^i, \hat{y^i}]}(-w^{i\prime\prime})$$

Then, for every s, $-w^{i\prime\prime}(s) \leq M^i$. Since $(s-y) \geq 0$, then

$$-w^{i\prime\prime}(s)(s-y) \le M^i(s-y)$$

Integrating both sides

$$\begin{aligned}
-w^{i\prime\prime}(s)(s-y) &\le M^i(s-y)\\
\int_y^{\bar{y}^i} -w^{i\prime\prime}(s)(s-y)ds &\le M^i \int_y^{\bar{y}^i} (s-y)ds\\
w^i(\bar{y}^i) - w^i(y) &\le M^i \int_y^{\bar{y}^i} (s-y)ds\\
w^i(\bar{y}^i) - w^i(y) &\le M^i \int_y^{\bar{y}^i - y} u du\\
w^i(\bar{y}^i) - w^i(y) &\le M^i \frac{u^2}{2}\Big|_{\bar{y}^i - y}^{0}\\
w^i(\bar{y}^i) - w^i(y) &\le M^i \frac{(\bar{y}^i - y)^2}{2}\\
0 \le w^i(\bar{y}^i) - w^i(y) &\le \tilde{B}^i (y - \bar{y}^i)^2
\end{aligned}$$

The maximum is finite because $w^{i\prime\prime}$ is continuous and no species can be driven to extinction due to the regularity condition.

The next lemma shows the shadow-value deviation is also bounded:

Lemma 4.2 (Bounds for the fisherman's response)**.** *Let $Y^i(\lambda) = \theta^i/(p^i - \lambda)$ be the fisherman's escapement response under a quota. Suppose the realized shadow value is confined to a bounded interval $\lambda_t^F \in [\lambda_L, \lambda_u]$ with $\lambda_u < \min_i p^i$. Then there exist finite positive constants*

$$\underline{m} := \min_i \frac{\theta^i}{(p^i - \lambda_L)^2} > 0, \qquad \overline{m}^i := \frac{\theta^i}{(p^i - \lambda_u)^2} < \infty, \tag{63}$$

such that, for any two shadow values $\lambda, \lambda' \in [\lambda_L, \lambda_u]$,

$$\underline{m}\,|\lambda - \lambda'| \;\le\; \left|Y^i(\lambda) - Y^i(\lambda')\right| \;\le\; \overline{m}^i\,|\lambda - \lambda'|. \tag{64}$$

In particular, taking $\lambda = \lambda_t^F$, $\lambda' = \lambda^+$ and $y_t^i = Y^i(\lambda_t^F)$, $\bar{y}^i = Y^i(\lambda^+)$,

$$\underline{m}\,|\lambda_t^F - \lambda^+| \;\le\; |y_t^i - \bar{y}^i| \;\le\; \overline{m}^i\,|\lambda_t^F - \lambda^+|. \tag{65}$$

The response function has derivative

$$Y^{i\prime}(\lambda) = \frac{d}{d\lambda}\frac{\theta^i}{p^i - \lambda} = \frac{\theta^i}{(p^i - \lambda)^2} > 0,$$

so Y^i is strictly increasing. Differentiating once more, $Y^{i\prime\prime}(\lambda) = 2\theta^i/(p^i - \lambda)^3 > 0$ on $\lambda < p^i$, so $Y^{i\prime}$ is itself increasing in λ. Then, the response is flattest at the lower end λ_L and steepest at the upper end λ_u. Hence on $[\lambda_L, \lambda_u]$,

$$\frac{\theta^i}{(p^i - \lambda_L)^2} \;\le\; Y^{i\prime}(\lambda) \;\le\; \frac{\theta^i}{(p^i - \lambda_u)^2} = \overline{m}^i$$

Taking the minimum over species gives the floor $\underline{m}$ in (63). Both constants are finite and strictly positive because $\lambda_u < \min_i p^i$ avoids the denominator becoming 0.

By the mean value theorem, for any λ, λ' there is a point ζ^i between them with $Y^i(\lambda) - Y^i(\lambda') = Y^{i\prime}(\zeta^i)(\lambda - \lambda')$. Since $\zeta^i \in [\lambda_L, \lambda_u]$, its slope lies between the two bounds, which gives (64).